Toxicology

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Learning Objectives

1. Distinguish between the common clinical toxidromes associated with acute poisonings.
2. Describe the general management of a patient with an acute overdose.
3. Determine the best options for the management of selected toxins.
4. Develop a patient care plan for a patient presenting with an acute overdose.
5. Identify the adverse effects and monitoring of the patient who is poisoned.

Abbreviations in This Chapter

ABC	Airway, breathing, and circulation
ALT	Alanine aminotransferase
AST	Aspartate aminotransferase
BP	Blood pressure
BUN	Blood urea nitrogen
cAMP	Cyclic adenosine monophosphate
CB1	Cannabinoid-1
CB2	Cannabinoid-2
CIWA	Clinical Institute Withdrawal Assessment for Alcohol Scale
CK	Creatine kinase
CNS	Central nervous system
COPD	Chronic obstructive pulmonary disease
ECG	Electrocardiogram
ED	Emergency department
GABAa	γ-aminobutyric acid
GI	Gastrointestinal
HIET	Hyperinsulinemic euglycemic therapy
HR	Heart rate
ICU	Intensive care unit
INR	International normalized ratio
NAPQI	*N*-acetyl-*p*-benzoquinoneimine
NMDA	N-methyl-d-aspartate
RR	Respiratory rate
SBP	Systolic blood pressure
SCr	Serum creatinine
SSRI	Selective serotonin reuptake inhibitor
TCA	Tricyclic antidepressant

Self-Assessment Questions

Answers and explanations to these questions can be found at the end of this chapter.

1. A 38-year-old woman with type 2 diabetes is admitted for confusion and altered mental status. She has an active prescription for glipizide 10 mg by mouth twice daily, but she cannot respond to further questioning. Her vital signs are as follows: blood pressure (BP) 115/65 mm Hg, heart rate (HR) 68 beats/minute, respiratory rate (RR) 15 breaths/minute, and temperature 98.6°F (37°C). Her point-of-care glucose concentration is 45 mg/dL, and she is given 50 mL of 50% dextrose in water intravenously twice. Her follow-up point-of-care glucose concentration is 50 mg/dL after the first dose and 57 mg/dL after the second dose, with no improvement in symptoms. Which is the most appropriate intervention at this time?
   A. Dextrose.
   B. Glucagon.
   C. Octreotide.
   D. Sodium bicarbonate.

2. A 56-year-old man is admitted to the intensive care unit (ICU) after a β-blocker overdose. After administering 2 L of 0.9% sodium chloride and 3 g of calcium gluconate, his vital signs are as follows: BP 70/40 mm Hg, HR 52 beats/minute, and RR 22 breaths/minute. Which therapy is most appropriate at this time?
   A. Glucagon 5 mg intravenously.
   B. Atropine 2 mg intravenously.
   C. Insulin 0.1 unit/kg intravenously.
   D. Dopamine 2 mcg/kg/minute intravenously.

3. A 76-year-old woman is admitted to the emergency department (ED) with the chief concern of decreased mental status. Her vital signs are as follows: BP 118/72 mm Hg, HR 57 beats/minute, and RR 17 breaths/minute. She is experiencing nausea, but her physical examination is otherwise normal. Her husband is concerned that she may not be taking her medications properly. Given her presentation, which common toxidrome is the patient most likely experiencing?

A. Anticholinergic.
B. Cholinergic.
C. Opioid.
D. Sympathomimetic.

4. A 38-year-old woman is admitted to the ICU after a suspected overdose of risperidone. She was initially hypotensive, but she was stabilized after the administration of two 500-mL boluses of lactated Ringer solution. Her BP is now 118/77 mm Hg, HR 75 beats/minute, and RR 16 breaths/minute. A 12-lead electrocardiogram (ECG) shows QT prolongation (corrected QT interval [QTc] = 480 milliseconds), and her chemistry panel is significant for a bicarbonate of 24 mEq/L, potassium of 3.1 mEq/L, and magnesium of 1.8 mg/dL. Which intervention is most appropriate at this time?
   A. Potassium chloride 20 mEq every hour for two doses.
   B. Activated charcoal 50 g.
   C. Magnesium sulfate 2 g.
   D. Lorazepam 2 mg.

5. A 48-year-old man is admitted to the medical floor for community-acquired pneumonia. His medical history is significant for hypertension, hyperlipidemia, and chronic obstructive pulmonary disease (COPD), and he reports occasional alcohol use but no history of alcohol withdrawal. He is initiated on levofloxacin 750 mg intravenously daily and nebulizer treatments with albuterol and ipratropium. Twenty-four hours after admission, he is increasingly more confused and has nausea and vomiting. His vital signs are stable: BP 115/68 mm Hg, HR 122 beats/minute, RR 21 breaths/minute, and temperature 99.7°F (37.6°C). The team is concerned about possible alcohol withdrawal and asks for recommendations for initial therapy. Which is the most appropriate treatment for this patient?
   A. Lorazepam 2 mg intravenous push every 4 hours as needed according to the patient's Clinical Institute Withdrawal Assessment for Alcohol Scale (CIWA) score.
   B. Phenobarbital 65 mg by mouth every 8 hours as needed according to the patient's CIWA score.
   C. Propofol continuous infusion.
   D. Clonidine 0.1 mg by mouth every 12 hours.

6. A 57-year-old male patient on the medical floor is incorrectly administered a dose of methadone 40 mg by mouth that was written for the patient in the adjoining bed. Two hours later, the nurse finds him unresponsive with the following vital signs: BP 105/67 mm Hg, HR 61 beats/minute, RR 8 breaths/minute, and temperature 98.7°F (37.1°C). The nurse calls for the rapid response team, and, as the team pharmacist, you are asked for a recommendation. Which treatment is most appropriate at this time?
   A. Activated charcoal 50 g orally.
   B. Naloxone 0.04 mg intravenously.
   C. Whole bowel irrigation orally.
   D. 1 L of 0.9% sodium chloride intravenously.

Questions 7 and 8 pertain to the following case.
A 56-year-old female patient is admitted to the ED after an intentional overdose of 25 amlodipine 10-mg tablets. She is given activated charcoal 50 g, 2 L of 0.9% sodium chloride, and 3 g of calcium gluconate. Her current vital signs are as follows: BP 90/50 mm Hg, HR 107 beats/minute, RR 17 breaths/minute, and temperature 98.7°F (37.1°C). Serum chemistries are as follows: Na 141 mEq/L, K 2.5 mEq/L, Cl 101 mEq/L, HCO_3 24 mEq/L, blood urea nitrogen (BUN) 19 mg/dL, serum creatinine (SCr) 0.9 mg/dL, and glucose 215 mg/dL. The ED physician wants to initiate hyperinsulinemic euglycemic therapy (HIET).

7. Which is most appropriate to initiate or do first with respect to HIET?
   A. Give insulin 1 unit/kg intravenously.
   B. Give 50 mL of 50% dextrose in water intravenously.
   C. Warn the physician that full effects may take up to 30 minutes.
   D. Give 20 mEq of potassium chloride intravenously every hour for four doses.

8. The patient is not responding to HIET initiation. Her current infusion rate is 1 unit/kg/hour. Her BP remains low at 70/40 mm Hg, and her HR is now 58 beats/minute. Her repeat glucose is 186 mg/dL. Which is most appropriate to initiate at this time?
   A. Discontinue HIET, and initiate norepinephrine.
   B. Continue HIET, and increase the insulin infusion rate and initiate norepinephrine.
   C. Continue HIET, and initiate epinephrine.
   D. Discontinue HIET, and begin intravenous lipid therapy.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1, 2, 3, 6
   b. Task B: 3, 4
2. Domain 2: Therapeutics and Managment
   a. Task B: 2, 3, 4, 9, 11, 15
   b. Task C: 1, 2, 3, 4
3. Domain 3: Professional Practice
   a. Task A: 1
   b. Task B: 1, 4

I. EPIDEMIOLOGY

A. Population based: The American Association of Poison Control Centers releases an annual report based on all the cases submitted by the 55 regional poison centers to the National Poison Data System to help clinicians stay abreast of the changing landscape of exposures (Clin Toxicol 2023;61:717-939).
   1. In 2022, 2,064,875 human exposures were reported. Fatalities were reported in 1491 cases (Table 1).
   2. The most common site of exposure was a residence (90.7%), followed by workplace (1.93%) and other's residence (1.81%).
   3. Most of the reported cases (1,159,170) occurred in children, defined in the report as younger than 20 years. To add perspective, 968,189 cases were reported in children 12 years and younger.
   4. The most common reasons associated with these exposures were unintentional (76.4%), intentional (18.9%), and adverse reactions (2.7%). Of note, therapeutic errors accounted for 331,539 (16.1%) of all cases. The scenarios reported for therapeutic errors included inadvertent double dosing (30.8%), incorrect dose (16.4%), incorrect medication administered or taken (15%), doses administered too close in time (11.1%), and inadvertent exposure to another person's medication (8.5%).
   5. Routes of exposure included ingestion (82.7%), inhalation/nasal (7.6%), dermal (6.9%), and ocular (4.1%). Of the 1491 exposure-related fatalities, most were by ingestion (83.4%), followed by inhalation/nasal (7.3%) and parenteral (2.7%).

B. Management based
   1. Only 30.6% of exposures were managed in a health care facility, with 3.4% managed in critical care units.
   2. Gastric decontamination was used in about half of all exposures (49.9%); however, antidotes were given in only 12.4% of the exposures. The most common gastric decontamination strategy was the use of activated charcoal (1.6% of total exposures), followed by other emetic agents (0.9%), cathartics (0.1%), whole bowel irrigation (0.05%), and gastric lavage (0.02%).

Table 1. Top 5 Most Common Medication-Related Toxic Exposures in 2022

Most Common Medication-Related Toxic Exposures	% of Total Cases
All human exposures	
Analgesics	11.51
Antidepressants	5.61
Antihistamines	4.81
Cardiovascular drugs	4.74
Sedatives/hypnotics/antipsychotics	4.60
Adult exposures (≥ 20 yr of age)	
Analgesics	11.09
Sedatives/hypnotics/antipsychotics	7.37
Antidepressants	7.30
Cardiovascular drugs	7.06
Alcohols	4.39
Anticonvulsants	3.81
Pediatric exposures (< 6 yr of age)	
Analgesics	9.54
Dietary/herbal/homeopathic supplements	6.65
Antihistamines	5.03
Vitamins	4.85
Topical preparations	4.06

Information from: Gummin DD, Mowry JB, Beuhler MC, et al. 2022 annual report of the American Association of Poison Control Centers' National Poison Data System (NPDS): 40th annual report. Clin Toxicol 2023;61:717-939.

II. EMERGENCY EVALUATION AND MANAGEMENT

A. The primary treatment strategy for managing a toxic exposure should focus on stabilizing the patient, with an emphasis on airway, breathing, and circulation (ABC). The most common factor contributing to death from a poisoning or drug overdose is the loss of the protective reflexes of the airway secondary to flaccid tongue, aspiration of gastric contents into the lungs, or respiratory compromise including arrest. Emergent endotracheal intubation should be performed when necessary. Vital signs (HR, RR, BP, temperature, and oxygen saturation) and changes in mental status should be closely monitored. After stabilization, the DEFG approach can be considered:
   D: Decontamination
   E: Enhanced elimination
   F: Focused antidote therapy
   G: Get help from a poison control center or toxicologist

B. Supportive care should be based on specific patient symptoms and may include the administration of intravenous fluids, supplemental oxygen, and advanced airway management. Other potential complications should be assessed, such as presence of rhabdomyolysis, rigidity, or dystonia. Additional tests, such as a 12-lead ECG, chest radiograph, or electroencephalogram may be required. In addition, essential laboratory tests should be conducted and assessed for the presence of an osmolar gap, anion gap acidosis, hyper/hypoglycemia, hyper/hyponatremia, hyper/hypokalemia, renal failure, and liver failure.
   1. Use of "coma cocktail" preparations is controversial and therefore not routinely recommended because they should not replace or substitute for a thorough analysis of the patient (JEMS 2002;27:54-60). Formulations vary, but they typically contain one or more of the following: dextrose, thiamine, and naloxone. The following text presents an overview of the common additives, a rationale for use, and potential controversies.
      a. Dextrose 50%, 12.5–25 g (25–50 mL) intravenously is administered to treat hypoglycemia; it is recommended to perform point-of-care blood glucose testing to confirm before administration.
      b. Thiamine 100 mg is administered intravenously to prevent Wernicke encephalopathy; often under-recognized, several doses of high-dose parenteral thiamine (e.g., 500 mg) concurrent with or immediately following intravenous dextrose before intravenous dextrose are typically required to effectively treat (J Emerg Med 2012;42:488-94).
      c. Naloxone 0.04–2 mg IV is administered in a stepwise titration to reverse respiratory depression secondary to opiate overdose.

C. Ingestions
   1. A thorough physical examination should be performed.
   2. A medication history and reconciliation should be done, including all prescription medications, over-the-counter agents, illicit substance use, and herbal products.
   3. The history of the ingestion should be determined, if possible, including the following elements (Ann Emerg Med 1999;33:735-56):
      a. Timing and route of the exposure, the possible agents involved and their strengths and amounts, and the potential intent of the patient
      b. History from the prehospital care providers, family members, or other patient advocates
      c. Onset and progression of any symptoms
   4. Some providers advocate for the use of toxidromes, which are a collection of symptoms that occur with particular classes of toxic agents. Toxidromes may help identify the toxic agent and assist in care by helping providers anticipate additional symptoms. Although they may be very useful in the care of an acute poisoning, they should be used with caution because some symptoms may overlap with other classes of toxins, may be impacted by comorbid conditions, or may be absent altogether.

5. Common Toxidromes and Presentation (Chest 2011;140:795-806) (Crit Care Clin 2012;28:180-198)
   a. Anticholinergic
      i. Mechanism of toxicity is through competitive antagonism of the effects of acetylcholine at peripheral muscarinic receptors and central receptors.
      ii. Signs and symptoms include altered mental status, delirium, hallucinations, mumbled speech, dry mucous membranes, mydriasis, anhidrosis, flushing, hyperthermia, tachycardia, hypoactive bowel sounds, and urinary retention.
      iii. Common drugs that have anticholinergic activity include antihistamines, antipsychotics, tricyclic antidepressants, and skeletal muscle relaxants.
   b. Cholinergic
      i. Mechanism of toxicity is inhibition of acetylcholinesterase causing accumulation of acetylcholine ultimately resulting in overstimulation of muscarinic and nicotinic receptors.
      ii. Signs and symptoms include confusion, central nervous system (CNS) depression, miosis, wet mucous membranes, salivation, lacrimation, diaphoresis, emesis, urination, diarrhea, muscle weakness/twitching, bronchospasm, bronchorrhea, hypertension, tachycardia, and bradycardia.
      iii. Common drugs that have cholinergic activity include organophosphates, nerve agents, nicotine, pilocarpine, and physostigmine.
   c. Opioid
      i. Mechanism of toxicity is stimulation of opioid receptors causing a decrease in autonomic activity.
      ii. Signs and symptoms include sedation, miosis, decreased bowel sounds (ileus), respiratory depression, hypotension, and bradycardia.
      iii. Common drugs that have opioid activity include heroin, morphine, codeine, synthetic opioids, loperamide, and dextromethorphan (in large quantities).
   d. Sympathomimetic
      i. Mechanism of toxicity is through an increase in sympathetic tone through release of cathecholamines, inhibition of reuptake, by direct receptor stimulation, and alterations in neurotransmitter metabolism.
      ii. Signs and symptoms include agitation, delirium, mydriasis, diaphoresis, myoclonus, hyperthermia, hypertension, and tachycardia.
      iii. Common drugs that have sympathomimetic activity include cocaine, methamphetamine, pseudoephedrine, and caffeine.
6. Drug screens are used in acute toxic ingestions, the most common of which is the qualitative urine screen. This method tests for the presence of a substance, but it cannot detect the amount of substance present. If a toxin is known, a quantitative drug screen may be used to confirm the exact amount present. Although urine drug screens may vary by institution, they may include amphetamines, barbiturates, benzodiazepines, cocaine, MDMA (ecstasy), methamphetamines, opiates, THC (marijuana), and tricyclic antidepressants (TCAs). Urine screens are not considered comprehensive; therefore, the presence of additional agents should be tested for (e.g., acetaminophen, salicylates). However, comprehensive drug screens using gas chromatography and liquid chromatography together with mass spectrometry can be used in patients presenting with severe or unexplained toxicity.
   a. A negative screen does not exclude the presence of a toxic substance, especially if the presumed agent is not present on the screen. Many agents are not identified by their designated screen; this is especially an issue with standard amphetamine, benzodiazepine, and opiate screens.

b. A positive test also does not necessarily confirm the diagnosis because another agent may be present but at concentrations below a detectable threshold. In addition, a positive test does not indicate that the patient is intoxicated on the particular substance (e.g., cocaine is positive for 3 days; however, its effects last only a few hours) or that the agent ingested is the exact agent that is screened (e.g., bupropion causes a false-positive amphetamine screen) (Am J Health Syst Pharm 2010;67:1344-50).

Patient Case

1. A 53-year-old man (height 74 inches, weight 97 kg [215 lb]) arrives in the ED confused and disoriented. He cannot provide any information about his condition or medical history. Vital signs are as follows: BP 85/50 mm Hg, HR 120 beats/minute, RR 28 breaths/minute, and temperature 99.2°F (37.3°C). On physical examination, an unmarked pill bottle is found in his pocket. Two tablets remain, and a possible drug overdose is suspected. Which is most appropriate to do first for this patient?
   A. Send a quantitative urine drug screen.
   B. Stabilize the patient's ABC.
   C. Order a coma cocktail.
   D. Try to identify the tablets in a drug database.

III. GASTRIC DECONTAMINATION/ENHANCED ELIMINATION

A. Many strategies for gastric decontamination are used to try to remove toxins or prevent further absorption. No particular strategy is preferred to another; each has certain advantages and disadvantages, and the risks and benefits must be considered before use. Consensus statements from the American Academy of Clinical Toxicology and the European Association of Poisons Centres and Clinical Toxicologists recommend against the routine use of any decontamination strategy but suggest that these strategies play a role in individualized care after a poison exposure (Clin Toxicol 2013;51:127). Table 2 lists the common dosing strategies for general decontamination and enhanced elimination.

B. Ipecac
   1. Ipecac is no longer manufactured in the United States since 2010 and is no longer recommended because of concerns for safety and ability to improve outcomes for patients who have been poisoned.
   2. Mechanism of action is to induce vomiting through irritation of the gastric mucosa and stimulation of the chemoreceptor trigger zone in the medulla.
   3. Guidelines recommend ipecac (if available) to be given only under a specific recommendation from a poison control center, ED physician, or other qualified medical personnel when all the following conditions are met (Clin Toxicol 2013;51:134-9; Clin Toxicol 2005;43:1-10):
      a. No specific contraindication exists for use.
      b. There is a substantial risk of serious toxicity of the toxin to the patient.
      c. No alternatives are available or considered effective to reduce the toxin absorption.
      d. A delay of more than 1 hour is expected before arrival to a medical facility.
      e. Use of ipecac will not adversely affect a more definitive treatment option.

C. Gastric Lavage
   1. Gastric lavage is performed by inserting a larger-bore orogastric or nasogastric catheter tube (36–40 French for adults and 24–28 French for children) with several holes at the distal end into the stomach. Aliquots of warmed tap water (200–300 mL) are then instilled until there is clearing of aspirated fluid.
   2. The efficacy is highly variable and diminishes over time; therefore, it is optimal to perform within 60 minutes of ingestion.
   3. Guidelines emphasize that gastric lavage has not been proven to decrease the severity of illness, improve recovery times, or improve outcomes (Clin Toxicol 2013;51:140-6).
   4. Should be considered only for life-threatening ingestions when it can be safely performed within 30–60 minutes of ingestion.
   5. Even in life-threatening overdoses, it may not be beneficial. Gastric lavage should not be performed routinely, if at all, for treating the patient who is poisoned. In the rare situation when it might be appropriate, clinicians should consider treatment with activated charcoal or observation and supportive care in place of gastric lavage (Clin Toxicol 2013;51:140-6).
   6. Contraindications for gastric lavage include patients with craniofacial abnormalities, concomitant head trauma, unprotected airway, increased risk of aspiration, those at risk of GI hemorrhage or perforation and caustic ingestion such as acids (e.g., boric acid) and alkalis (e.g., dishwasher detergents) because of the risk of exacerbating any esophageal or gastric injury. Patients with decreased consciousness require endotracheal or nasal intubation before the procedure.
   7. Complications associated with gastric lavage include aspiration, laryngospasm, perforation of the esophagus or stomach, arrhythmias, fluid imbalance, hyponatremia, and small conjunctival hemorrhages.

D. Cathartics
   1. Used to reduce the transit time of toxins and hence absorption, as well as in combination with activated charcoal to decrease constipating effects
   2. Cathartics have conflicting data regarding decreased transit time in the GI tract and have no data to support improved patient outcomes (J Toxicol Clin Toxicol 2004;42:243-53).
   3. Cathartic use is not recommended; if used, it should be limited to a single dose and not be used as monotherapy.
   4. Contraindications to cathartic use include absence of bowel sounds, recent GI surgery, intestinal perforation or obstruction, hypotension, electrolyte disturbances, and renal insufficiency (for magnesium-based cathartics).
   5. Complications include nausea, dehydration, hypotension, and magnesium imbalances.

E. Activated Charcoal
   1. Activated charcoal is an adsorbent that works by binding the toxin throughout the GI tract to reduce systemic absorption. Although activated charcoal binds most substances, Table 3 lists the agents for which activated charcoal is NOT recommended.
      a. Acids and alkalis should be avoided because charcoal may cause vomiting, which can be damaging in these ingestions. The black color and thickness of the activated charcoal may also cause discoloration of the stomach lining and therefore interfere with endoscopy.
      b. Alcohols bind poorly; therefore, large doses are needed, which are difficult to ingest.
      c. Cyanide will bind, but not with as much activity as other substances. Because the toxic dose of cyanide is so small, normal doses of activated charcoal may be ineffective.
      d. Hydrocarbons may lead to a significant risk of aspiration.
   2. It is optimal to administer activated charcoal within 60 minutes of the toxin ingestion to maximize efficacy.

3. The use of a cathartic (e.g., sorbitol) in combination with activated charcoal is not recommended.
4. If significant nausea occurs, it is recommended to administer an antiemetic. When choosing an antiemetic, potential drug and symptom interactions should be considered as well.
5. Complications include aspiration, accidental administration into the lung, emesis, constipation, and gastric obstruction.
6. Contraindications include an unconscious state or an inability to otherwise protect the airway without endotracheal intubation, ileus or intestinal obstruction, late presentation (more than 2 hours), and recent GI surgery.
7. Multidose activated charcoal is a method described to enhance the elimination of certain toxins. It is not more effective in reducing morbidity or mortality than single-dose charcoal, but it may be administered to enhance elimination in life-threatening ingestions caused by medications that undergo significant enterohepatic recirculation with active enterohepatic metabolites (J Toxicol Clin Toxicol 1999;37:731-51).

F. Whole Bowel Irrigation
1. Whole bowel irrigation is a strategy for cleansing the bowel to remove potential toxins by administering an osmotic polyethylene glycol solution.
2. Not recommended for routine use, but may be useful in potentially life-threatening ingestions of medications with long half-lives, sustained-release dosage forms, or enteric-coated formulations. Specifically useful for certain toxic substances not adsorbed by activated charcoal (e.g., lithium and iron). May also be beneficial for iron overdoses and for "packers" of illicit substances.
3. Concurrent administration of activated charcoal and whole bowel irrigation may decrease the efficacy of charcoal.
4. Complications of the polyethylene glycol electrolyte solutions include anaphylaxis, angioedema of the lips, aspiration, Mallory-Weiss tear, and esophageal perforation.
5. Contraindications include bowel obstruction, perforation, ileus, and use in patients with recent bowel surgery. A kidney-ureter-bladder radiograph may be used to rule out these contraindications.

G. Urine Alkalinization
1. Urine alkalinization is a strategy to improve the elimination of toxins by increasing the urine pH to levels of 7.5 or greater via administration of sodium bicarbonate or sodium acetate (J Toxicol Clin Toxicol 2004;42:1-26).
2. Specific substances that may benefit from this strategy include salicylates, phenobarbital, chlorpropamide, and other weak acids with intrinsic urinary clearance.
3. Contraindications include acute and chronic renal failure and preexisting heart failure owing to the volume of fluid required for this treatment strategy.
4. Complications include hypokalemia, hypernatremia, hypocalcemia, cerebral vasoconstriction, and coronary vasoconstriction.
5. To administer urine alkalinization, it is recommended to check baseline blood chemistries, electrolyte values, and an arterial blood gas, as well as to correct any fluid or electrolyte deficits (especially potassium because alkalemia will push potassium intracellularly). Hypokalemia will make it impossible to get the urine alkaline because of the K+-H+ exchange in the kidneys, which will excrete H+ into urine if K+ is low.
6. Guideline-recommended monitoring includes urine pH every 15–30 minutes (every 30–60 minutes is more accepted in clinical practice) until the goal pH level of 7.5–8.5 is achieved, followed by every hour; serum potassium concentrations, central venous pressure, and arterial blood gases should be measured hourly.

Table 2. Common Dosage Strategies for General Decontamination and Enhanced Elimination

Decontamination/ Elimination Strategy	Pediatric Dosing	Adult Dosing
Gastric lavage[a]	10-mL/kg aliquots, followed by return of an equal amount	200- to 300-mL aliquots, followed by return of an equal amount
Cathartics		
Magnesium citrate:	4 mL/kg	240 mL
Sorbitol:	4.3 mL/kg (35% solution)	1–2 mL/kg (70% solution)
Activated charcoal		
Single dose	Up to 1 yr of age: 0.5–1 g/kg (usually 10–25 g) 1–12 years: 0.5–1 g/kg (usually 25–50 g)	> 12 years and adults: 25–100 g (doses > 50 g may induce vomiting)[b]
Multidose	0.5–1 g/kg (25–50 g), followed by 0.25–0.5 g/kg (10–25 g) every 4 hr	50 g, followed by 25–50 g every 4 hr
Whole bowel irrigation[c]	9 mo to 6 yr: 500 mL/hr 6–12 yr: 1000 mL/hr	> 12 yr and adults: Goal is 2000 mL/hr (initiated at 500 mL/hr and doubled every 30 min)
Urine alkalinization[d]	25–50 mEq intravenously for 1 hr	250 mEq intravenously for 1 hr

[a]Sterile water or 0.9% sodium chloride; may repeat until the return fluid is clear and absent of particulate matter.

[b]Upper limit may vary depending on the capacity of the stomach.

[c]Polyethylene glycol electrolyte lavage solutions; dose until the rectal effluent is clear or the desired effect has been achieved.

[d]Sodium bicarbonate in dextrose 5% in water additional boluses can be given hourly (or begin a continuous infusion at this hourly rate) to maintain a urine pH of 7.5–8.5.

Table 3. Agents for Which Activated Charcoal Is NOT Recommended

Substance	Examples
Acids	Boric acid, mineral acids
Alcohols	Ethanol, ethylene glycol, methanol
Alkalis	Bleach, cleaning solutions, dishwasher detergents, lye
Carbamates	Insecticides, neostigmine, physostigmine
Cyanide	Cyanogen chloride, hydrogen cyanide, potassium cyanide, sodium cyanide
Hydrocarbons	Gasoline, kerosene, petroleum oils
Metals	Arsenic, iron, lead, lithium, mercury
Organic solvents	Acetic acid, acetone, ethylene glycol, glycerin, toluene
Organophosphates	Antihelminthic drugs (trichlorfon), insecticides (malathion, parathion), herbicides

IV. ACETAMINOPHEN

A. Background
   1. Acetaminophen is consistently one of the most common toxic drug exposures.
   2. Accounted for 66,710 exposures (as a single agent) and resulted in 167 deaths in 2022.
   3. In general, acute doses of 150 mg/kg or 7.5 g in adults and 200 mg/kg in children are considered toxic. It is recommended that doses exceeding this threshold be managed in a health care facility.

4. The mechanism of toxicity is caused by the active metabolite *N*-acetyl-*p*-benzoquinoneimine (NAPQI), which can lead to oxidant cell injury, hepatic failure, and death.
5. Around 90% of acetaminophen undergoes phase II conjugation to glucuronide and sulfate conjugates that are excreted in the urine. An additional 2% is excreted unchanged in the urine. The remaining amount (8%–10%) is converted by cytochrome P450 (CYP2E1) to NAPQI. NAPQI is normally converted by glutathione to cysteine conjugates, which are renally excreted. In an overdose, the sulfation and glucuronidation pathways become saturated, leading to glutathione depletion and a subsequent buildup of NAPQI (Clin Liver Dis 2013;17:587-607).

B. Clinical Presentation – Four clinical phases are associated with acetaminophen toxicity (time intervals are estimated and may vary with individual patients).
1. Phase I occurs within the first 24 hours after ingestion. Patients may present with minimal or no signs of distress. Potential signs and symptoms include nausea, vomiting, diaphoresis, and anorexia.
2. Phase II occurs 24–48 hours after exposure and is marked by initial damage to the hepatocytes. Patients may present with right upper quadrant pain, increases in liver transaminases, elevated total bilirubin concentrations, and prolonged prothrombin time.
3. Phase III occurs 72–96 hours after initial exposure and is the peak of the hepatotoxic effects. Patients may present with lactic acidosis, acute renal failure, acute pancreatitis, and fulminant hepatic failure, as evidenced by jaundice, extensive coagulopathies, hypoglycemia, and hepatic encephalopathy.
4. Phase IV occurs about 1 week after exposure and marks the recovery phase if the patient survives phase III.

C. Treatment
1. The goal of treatment is to prevent the development of hepatic toxicity and reduce mortality.
2. Gastric decontamination with a single dose of activated charcoal can be considered if the patient presents within the first hour after exposure, is not vomiting, and has no alterations in mental status. However, with the availability of highly efficacious antidotal therapy, the risks often outweigh the benefits.
3. Antidote therapy is recommended with acetylcysteine. The mechanism of action for acetylcysteine is to increase the synthesis and bioavailability of glutathione, substituting for glutathione by binding to the reduced sulfur group of NAPQI, and supplying a substrate for sulfation, thereby increasing nontoxic metabolism. Additional mechanisms of action have also been proposed, including its facilitation of reactive oxygen and nitrogen species scavenging.
4. Guidelines suggest that acetylcysteine treatment be administered to patients within the first 8 hours of exposure if they can be stratified as being at possible or probable risk of hepatotoxicity by the Rumack-Matthew nomogram (Figure 1). If patients cannot be stratified because of unknown time of ingestion, they should receive acetylcysteine if any of the following conditions apply: significantly elevated alanine aminotransferase (ALT) concentration, serum acetaminophen concentrations greater than 20 mcg/mL, or history of chronic ingestions exceeding 4 g/day with an elevated serum ALT concentration and/or presenting more than 24 hours post-ingestion with evidence of hepatotoxicity (Ann Emerg Med 2007;50:292-313).
    a. This includes patients presenting more than 24 hours postingestion with evidence of hepatotoxicity.
    b. Limitations to use of the Rumack-Matthew nomogram include the following (Ann Emerg Med 2007;50:292-313):
        i. Presentation more than 24 hours postingestion
        ii. An unknown or unreliable history of ingestion
        iii. Overdoses with extended-release formulations
        iv. Chronic or repeated supratherapeutic ingestions
        v. Patients with preexisting hepatic disease, chronic alcohol use, or concurrent medications metabolized by the CYP system or that alter gastric motility.

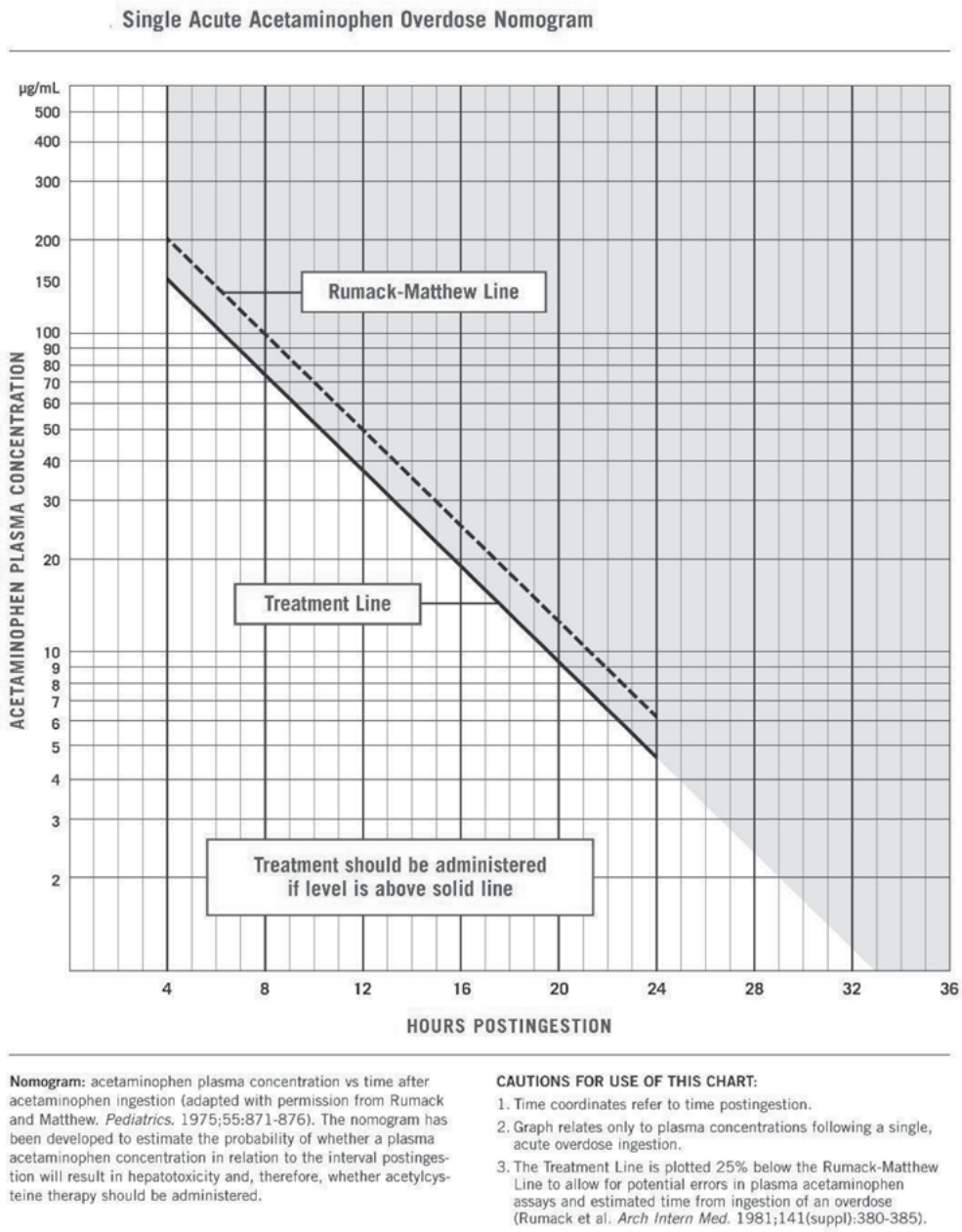


Figure 1. Rumack-Matthew nomogram.

Reprinted from: Tylenol for Healthcare Professionals. Guidelines for the Management of Acetaminophen Overdose. McNeil Consumer & Specialty Pharmaceuticals. Available at www.tylenolprofessional.com/guidelines-and-studies.html.

5. Intravenous acetylcysteine is advantageous because of its decreased overall administration time (21 hours vs. 72 hours for oral) and minimal GI adverse effects. If intravenous acetylcysteine formulation is not available and cannot be obtained in a timely fashion, poison control centers can be contacted for instructions on compounding the inhalational acetylcysteine formulation for intravenous use. It is not recommended to use this strategy except for emergencies.

6. Oral acetylcysteine is dosed for 18 total doses; doses may be repeated if emesis occurs within 1 hour of a dose. Two oral formulations exist. One preparation is an effervescent tablet that must be dissolved in water before administration, and the other is a liquid preparation that may be diluted in juice or carbonated beverages to improve palatability. Antiemetics may be administered if significant nausea or vomiting occurs. Table 4 discusses the dosing strategies for oral and intravenous administration of acetylcysteine.
    a. For exceptionally large ingestions (e.g., over 30 g), some studies have recommended alternative approaches:
        i. An increased dose of acetylcysteine (Med J Aust 2020;212:175-83).
        ii. Given that acetaminophen is dialyzable, intermittent hemodialysis is an option (Clin Toxicol 2014;52:856-67). This approach would also require an increase in the acetylcysteine dose (i.e., 2-fold) due to its concurrent increased clearance (Clin Toxicol 2016;54:519-22).
7. Although treatment guidelines recommend 18 total doses of acetylcysteine administered throughout 72 hours for oral acetylcysteine therapy and 21 hours of the intravenous infusion of acetylcysteine, many poison control centers recommend early discontinuation if the following conditions are met (or prolonged therapy if they are not met) (Ann Emerg Med 2007;50:280-1):
    a. Serum acetaminophen concentrations are undetectable or less than 10 mcg/mL.
    b. ALT concentrations are normal (60 IU/L or less) or improving. Some clinicians also advocate for an international normalized ratio (INR) of 1.3 or less.
    c. The patient is clinically improved.
8. Adverse effects (intravenous): Anaphylactoid reactions (rash, urticarial, pruritus) that are often infusion rate-associated, hyponatremia, hypervolemia, seizures (pediatric patients with unadjusted volume)
9. Adverse effects (oral): Nausea, vomiting, anaphylactoid reactions (rare)
10. Patients experiencing mild anaphylactoid reactions (rash, pruritus, flushing) can be effectively treated with diphenhydramine, and acetylcysteine therapy can be resumed.
11. Medication errors may occur because of the complex dosing regimens (Ann Pharmacother 2008;42:766-70). Common errors include delays in therapy, incorrect dosages, and incorrect infusion rates. To mitigate these errors, some institutions have developed a two-step intravenous regimen (Toxicol Commun 2018;2:81-4).

Table 4. Acetylcysteine Dosage

Route	Dose
Oral	**Loading dose:** 140 mg/kg **Maintenance doses:** 70 mg/kg every 4 hr for a total of 17 doses (72 hr)
Intravenous	**Loading dose:** 150 mg/kg (max 15 g)[a] in 200 mL of 5% dextrose in water infused for 60 min **Maintenance dose:** 50 mg/kg (max 5 g)[a] in 500 mL of 5% dextrose in water infused for 4 hr followed by 100 mg/kg (max 10 g)[a,b] in 1000 mL of 5% dextrose in water infused for 16 hr Patients weighing < 40 kg require reduced volume administration

[a] Dose limits are provided by the manufacturer; however, small retrospective studies have shown that acetylcysteine is well tolerated when using actual body weight for patients > 100 kg and exceeding the recommended max doses (Am J Ther 2016;23:e714-9.)(Am J Ther 2014;21:159-63).

[b]Higher ongoing infusion rates (e.g., 200 mg/kg over 16 hr) may be required for massive paracetamol ingestions (i.e., initial concentration that is double the standard nomogram line) and a clinical toxicologist should be consulted (Chiew AL, Reith D, Pomerleau A, et al. Updated guidelines for the management of paracetamol poisoning in Australia and New Zealand. Med J Aust 2020;212:175-183. doi: 10.5694/mja2.50428).

D. Monitoring
   1. Patients should be monitored for improvement in vital signs and mental status.
   2. The following laboratory values should be monitored periodically for improvement as well as for potential worsening.
      a. ALT, aspartate aminotransferase (AST), total bilirubin, INR, and prothrombin time
      b. Acetylcysteine may cause a dose-dependent decrease in prothrombin time, although this has no clinical effect on coagulation
      c. BUN and SCr
      d. Serum electrolytes
      e. Fulminant hepatic failure: Serum bicarbonate, serum sodium, serum lactate, arterial blood gas, serum glucose, and ammonia concentrations

Patient Case

Questions 2 and 3 pertain to the following case.
A 42-year-old woman (height 66 inches, weight 79.2 kg [176 lb]) presents to the ED with the chief concern of flu-like symptoms. Her symptoms include headache, congestion, severe nausea and vomiting, abdominal pain, and some confusion. She has been taking acetaminophen 500-mg caplets as needed for her symptoms, but she just ran out of the bottle she purchased yesterday. On presentation, she is alert and oriented. Her vital signs are as follows: BP 135/90 mm Hg, HR 83 beats/minute, RR 18 breaths/minute, and temperature 101.8°F (38.8°C). An acetaminophen concentration on admission was 100 mcg/mL, AST 560 IU/L, and ALT 310 IU/L. The physician wants to begin general management.

2. Which general management strategy is most indicated for this patient?
   A. 10% magnesium citrate 240 mL per tube once
   B. Continued stabilization of the patient and good supportive care
   C. Gastric lavage
   D. Charcoal 50 g once

3. Which is the most appropriate treatment for her acetaminophen toxicity?
   A. Give acetylcysteine 11,200 mg oral bolus, followed by 5600 mg orally every 4 hours for 17 doses.
   B. Give acetylcysteine 11,200 mg intravenous bolus, followed by 5600 mg intravenously every 4 hours for 12 doses.
   C. Give acetylcysteine 12,000 mg intravenously over 1 hour, followed by 4000 mg intravenously over 4 hours; then 8000 mg intravenously over 16 hours.
   D. Acetylcysteine therapy is not indicated in this patient.

V. SALICYLATES

A. Background
   1. Salicylates as a single agent (not in combination with other agents) accounted for 6595 overdoses and 18 deaths in 2022. These numbers, which include overdoses of both adult and pediatric formulations of acetylsalicylic acid, are often underreported because these products are not typically recognized as a potential cause.
   2. However, because of the development of child-resistant containers, aspirin overdose causing accidental death in children has reduced drastically. (Lancet 1977;2: 289-90)

B. Clinical Presentation
   1. The mechanism of toxicity for salicylates is through the interference with aerobic metabolism owing to the uncoupling of mitochondrial oxidative phosphorylation, leading to increases in anaerobic metabolism, which causes a significant lactic acidosis (Emerg Med Clin North Am 2007;25:333-46). This also leads to hypoglycemia because of glycogen depletion, gluconeogenesis, and catabolism of proteins and free fatty acids. Salicylates also directly stimulate the respiratory center, leading to hyperventilation and respiratory alkalosis. Secondary complications from hyperventilation include dehydration and compensatory metabolic acidosis.
   2. Salicylates are readily absorbed in the stomach and small intestine and are then conjugated with glycine in the liver to the active component, salicylic acid. In overdoses, the liver cannot metabolize the excess drug, and most is then excreted unchanged by the kidneys (Postgrad Med 2007;121:162-8).
   3. The most common clinical symptoms associated with a salicylate overdose are hyperventilation (causing respiratory alkalosis), tinnitus, and GI irritation. Symptoms may vary depending on the serum salicylate concentration; however, these may be low to normal early in the presentation (Postgrad Med 2007;121:162-8):
      a. Serum concentration less than 30 mg/dL: Asymptomatic
      b. Serum concentration 15–30 mg/dL: Therapeutic concentrations
      c. Serum concentration 30–50 mg/dL: Hyperventilation, nausea, vomiting, tinnitus, dizziness
      d. Serum concentration 50–70 mg/dL: Tachypnea, fever, sweating, dehydration, listlessness
      e. Serum concentration greater than 70 mg/dL: Coma, seizures, hallucinations, stupor, cerebral edema, dysrhythmias, hypotension, oliguria, renal failure
   4. Acute salicylate toxicity is typically associated more with GI symptoms; chronic toxicity is more associated with CNS-type symptoms.
   5. Absorption may be delayed up to 36 hours because of gastric pylorospasm, bezoar formation (precipitate concretions because of poor solubility), or enteric-coated formulations; therefore, these ranges should be used with caution because they may not correlate with actual symptoms (Am J Emerg Med 2010;28:383-4).

C. Treatment
   1. There is no antidote for salicylate poisoning; the goals of therapy are to limit the additional absorption of salicylates and to provide supportive care.
   2. Maintain a patent airway and assist ventilation, if necessary. Ensure adequate ventilation to prevent respiratory acidosis; however, do not mechanically ventilate patients unless clinically necessary as this will interfere with the patient's ability to appropriately compensate and maintain pH.
   3. Gastric decontamination with a single dose of activated charcoal is recommended within 60 minutes of acute ingestions and if the patient is alert with the absence of vomiting. However, there may be a role for administration after this time frame, as well as multidose administration based on the characteristics of the ingestion.
   4. Administer intravenous crystalloid fluids at a rate of 10-20 mL/kg/hr for the first 2 hours to maintain a normal BP and a urine output of 1–1.5 mL/kg/hr.
   5. Administer intravenous glucose for hypoglycemia or significant neurologic symptoms.
   6. Urine alkalinization is recommended to enhance renal elimination and increase the glomerular filtration rate.
      a. The following administration strategies have been described in the literature:
         i. Administer 250 mL of sodium bicarbonate 8.4% over 1 hour; then administer additional 50-mL boluses as needed to maintain a goal urine pH range of 7.5–8.5.
         ii. Administer a continuous infusion of 150 mL of sodium bicarbonate 8.4% in 1 L of 5% dextrose in water at 2–3 mL/kg/hour to maintain a urine output of 1–2 mL/kg/hour. Do not allow the serum pH to fall below 7.4.

   b. Oral bicarbonate is not recommended because it may enhance salicylate absorption by accelerating tablet dissolution.
   c. Discontinue bicarbonate once the serum salicylate concentrations are less than 30 mg/dL or there is a resolution of clinical symptoms.
7. Alkaline diuresis will increase potassium secretion, and hypokalemia will make it more challenging to raise the urine pH. Adequate potassium concentrations should be aggressively maintained.
8. Consider hemodialysis for any of the following (Postgrad Med 2007;121:162-8):
   a. Acute renal insufficiency
   b. End-organ damage (severe pulmonary edema, seizures, rhabdomyolysis)
   c. Altered mental status
   d. Deterioration of clinical status
   e. Severe acid-base disturbances
   f. If hemodialysis is not effective, hemoperfusion and continuous renal replacement therapies may be considered (Ann Emerg Med 2015;66:165-81).

D. Monitoring
1. Patients should be monitored for up to 24 hours because of the possibility of delayed or impaired absorption.
2. Monitor RR, and support as needed; caution is advised if intubation is required because of a requirement for an increased minute ventilation (Am J Emerg Med 2010;28:383-4).
3. During urine alkalinization, monitor for signs and symptoms of fluid overload, hypernatremia, hypokalemia, hypocalcemia, and worsening alkalemia.

Patient Case

4. A 62-year-old man presents to the ED with the chief concern of nausea, tachypnea, and flu-like symptoms. He is alert and oriented and can communicate that his symptoms have been worsening for the past 2 days. He has been taking a combination cold product, which he thinks has helped. His medical history is significant for a stroke, for which he takes aspirin 325 mg by mouth daily, and hypertension, for which he takes amlodipine 5 mg by mouth daily. His vital signs are as follows: BP 135/82 mm Hg, HR 78 beats/minute, RR 29 breaths/minute, and temperature 100.2°F (37.9°C). Arterial blood gas results are as follows: pH 7.52, Pco_2 25, and HCO_3 20 mEq/L. A salicylate concentration is sent, which is 25 mg/dL. Which treatment management strategy is most indicated for this patient?
   A. Sodium chloride infusion
   B. Urgent endotracheal intubation
   C. Sodium bicarbonate infusion
   D. Hemodialysis

VI. OPIOIDS

A. Background
   1. Opioids as a single agent (not in combination with other agents) accounted for 20,392 overdoses and 346 deaths in 2022. The Centers for Disease Control and Prevention reports that almost 15,000 deaths are caused by prescription opioid painkillers annually.
   2. The most common agents associated with a toxicologic event were fentanyl (prescription and non-prescription), heroin, oxycodone, buprenorphine, and tramadol.
   3. The most common agents associated with a toxicologic death were fentanyl (prescription and non-prescription), heroin, oxycodone, methadone, and morphine.
   4. Opioids act at the mu, delta, and kappa opioid receptors, although mu is responsible for most of the opioids' clinical effects.

B. Clinical Presentation
   1. The most common clinical symptoms associated with opioid overdose are respiratory depression (defined as fewer than 12 breaths/minute), coma, miosis, and hypoactive bowel sounds.
   2. Additional findings may include stupor, hepatotoxicity, acute renal failure, rhabdomyolysis, compartment syndrome, hypothermia, and seizures (N Engl J Med 2012;367:146-55).
   3. Diagnostic workup is as follows: (Pharmacotherapy: A Pathophysiologic Approach, 9e. New York: McGraw-Hill, 2014):
      a. 12-lead ECG to assess for QT prolongation (methadone may cause QT prolongation leading to torsades de pointes).
      b. Arterial blood gas to assess for respiratory acidosis
      c. Standard chemistry panel for electrolyte and glucose abnormalities as well as Creatine kinase (CK), BUN, and SCr for signs of rhabdomyolysis
      d. Pulse oximetry

C. Treatment
   1. Stabilize the airway, provide supplemental bag-valve mask respirations if needed, and administer supplemental oxygen. Establish an airway, if needed, by endotracheal intubation.
   2. Administer intravenous crystalloid fluids to maintain BP.
   3. Gastric decontamination with a single dose of activated charcoal may be considered only if the patient presents within the first hour after exposure and is awake with an intact airway. Whole bowel irrigation can be considered for extended-release formulations or for "packers" of illicit substances (including ingestion of fentanyl patches).
   4. Antidote therapy (N Engl J Med 2012;367:146-55; Circulation 2023;148:e149-84):
      a. Naloxone is a competitive antagonist at the opioid receptor.
      b. The intravenous route is preferred, but naloxone is also effective through the endotracheal, intramuscular, intranasal, inhalational, intraosseous, or intrapulmonary route. However, it is important to note that the bioavailability and onset of action can vary greatly between different routes.
      c. Onset of action of intravenous naloxone is 1–2 minutes, with a duration of 30–120 minutes.
      d. Dosing may be affected by the specific opioid agent and dose, affinity for the mu-receptor, and patient weight. Goal of care is reversal of ventilatory depression, not necessarily reversal of neurologic depression, as this is more likely to precipitate acute opioid withdrawal syndrome.

e. Initial dose is 0.04 mg in adult patients and 0.1 mg/kg in pediatric patients; if no response, the dose is increased every 2–3 minutes to 0.5 mg, 2 mg, 4 mg, and 10 mg, followed by 15 mg. Large doses and frequent repeat administration of naloxone are necessary for patients thought to have had an overdose of heroin laced with carfentanil.
f. It is recommended that a continuous infusion be initiated at a dose of two-thirds the effective bolus dose per hour (0.04–4 mg/hour) for patients requiring subsequent naloxone doses to sustain effect secondary to exposure to long-acting opioids (e.g., methadone) or a sustained-release product (Ann Emerg Med 1986;15:566-70).
g. Intranasal administration appears to be safe and effective when the intravenous route is not available. The dose is administered by attaching a 2 mg/2 mL naloxone prefilled syringe to an atomizer and spraying an equal portion (1 mL) of the contents into each nostril. Commercial preparations contain up to 4 mg per actuation.
h. Adverse effects are rare and may be more related to a return of sympathetic response to opioid withdrawal. Nausea and vomiting are common and can lead to aspiration. If the situation allows, providing assisted manual ventilation prior to administering naloxone can potentiate the sympathetic response to opioid reversal (Anesth Analg 1988;67:730-6).
i. If no effect is seen at the higher naloxone doses, consider other causes such as co-ingestants or alternative agents.

D. Monitoring – Observe respiratory status and vital signs for a minimum of 4 hours after the last dose of naloxone or discontinuation of the continuous infusion. The duration of a naloxone continuous infusion will vary based on the quantity, pharmacokinetics, and pharmacodynamics of the opioid exposure, in addition to patient specific factors. Closely monitor for signs and symptoms of opioid withdrawal syndrome, such as anxiety, piloerection, heightened sensation to pain, abdominal cramps, diarrhea, and insomnia.

VII. LOPERAMIDE

A. Background
1. A nonprescription antidiarrheal medication with a 91% increase in reported abuse in 2010-2015 resulting in 15 deaths (Ann Emerg Med 2017;69:73-8). Loperamide as a single agent (not in combination with other agents) accounted for 695 overdoses and 1 death in 2022.
2. Relatively safe at therapeutic doses; however, can be fatal when high doses are ingested for its euphoric effects
3. Phenylpiperidine opioid that slows intestinal transit time by stimulating mu-opioid receptors in the GI tract and blocks intestinal calcium channels
4. Toxicity of loperamide involves blockade of sodium channels and potassium channels in the cardiac tissue, causing QT prolongation and QRS interval widening and leading to life-threatening dysrhythmias and cardiac death

B. Clinical Presentation
1. Common signs/symptoms include respiratory depression, nausea, vomiting, decreased level of consciousness, miosis, decreased bowel motility/ileus, palpitations, and syncope.
2. In severe toxicity, ECG findings are abnormal, including widened QRS interval and prolonged QT interval.

3. Diagnostic workup
    a. 12-lead ECG to assess for QT prolongation and QRS interval widening and development of ventricular dysrhythmias
    b. Arterial blood gas to monitor for respiratory acidosis secondary to respiratory depression
    c. Standard chemistry panel for electrolyte and glucose abnormalities, plus CK, BUN, and SCr for signs of rhabdomyolysis
    d. Pulse oximetry

C. Treatment
1. Stabilize the airway and provide supplemental oxygen, if needed. Establish an airway if patients cannot protect their airway or have significant respiratory depression.
2. Administer intravenous crystalloid fluids to maintain BP.
3. Gastric decontamination with a single dose of activated charcoal may be considered only if patients present within 2-4 hours after a large overdose and can protect their airway.
4. Antidote therapy (Ann Emerg Med 2017;pii: S0196-0644(17)30424-9.):
    a. Naloxone
        i. Should be administered in addition to supportive care in patients with respiratory depression who cannot protect their airway
        ii. The lowest effective dose should be used; repeat doses may be necessary secondary to loperamide's slow elimination.
    b. Sodium bicarbonate
        i. May be beneficial in patients with signs and symptoms of cardiac toxicity and QRS interval widening secondary to sodium channel blockade by loperamide
        ii. Sodium bicarbonate 1-2 mEq/kg intravenously should be administered concomitantly with magnesium and potassium chloride if electrolyte abnormalities exist.

D. Monitoring
1. Patients should be closely monitored for resolution of clinical symptoms, including respiratory status, vital signs, and return to baseline mental status.
2. Monitor serum electrolytes, blood glucose, and 12-lead ECG periodically.
3. Closely monitor for signs and symptoms of opioid withdrawal.

VIII. ALCOHOLS (METHANOL AND ETHYLENE GLYCOL)

A. Background
1. Alcohol poisonings (methanol and ethylene glycol) are not as common as poisonings with other substances, accounting for <1% of all cases in 2022 (National Poison Data System), but they can be serious and potentially fatal.
2. Methanol is commonly found in products such as windshield washer fluid, antifreeze, brake and carburetor fluids, and cooking products.
3. Ethylene glycol is commonly found in products such as antifreeze, de-icing solutions, refrigerants, and brake fluids.
4. Toxicity of both agents is caused by the breakdown to toxic metabolites by alcohol dehydrogenase and aldehyde dehydrogenase.

a. Methanol is converted to formaldehyde and then to formic acid, which results in an anion gap acidosis and ocular toxicity.
b. Ethylene glycol is converted to glycoaldehyde and then to glycolic acid, followed by glyoxylic acid, and, eventually, oxalic acid. Glycolic acid results in an anion gap acidosis and CNS toxicity. Oxalic acid results in CNS toxicity and renal toxicity because of the formation of calcium oxalate crystals.

B. Clinical Presentation
1. Common symptoms include inebriation, altered mental status, nausea, vomiting, hematemesis, nystagmus, and depressed reflexes. In rare cases of ethylene glycol toxicity, patients may present with tetany caused by hypocalcemia. Symptoms typically develop over the first 24 hours; however, they can be delayed for days in the setting of ethanol coingestion.
2. Early in therapy, an osmolar gap will be present, but this will diminish as the parent compound is metabolized.
   a. As the osmolar gap declines, the anion gap will rise, resulting in a significant metabolic acidosis.
   b. Calculations:
      i. Osmolar gap (OG):
      Measured osmolality - calculated osmolality (normal OG less than 10)
      ii. Calculated osmolality:
      (sodium x 2) + (glucose/18) + (BUN/2.8)
      iii. Calculated osmolality with ethanol ingestion:
      (sodium x 2) + (glucose/18) + (BUN/2.8) + (ethanol/4.6)
      iv. Calculated osmolality with methanol ingestion:
      (sodium x 2) + (glucose/18) + (BUN/2.8) + (methanol/3.2)
      v. Calculated osmolality with ethylene glycol ingestion:
      (sodium x 2) + (glucose/18) + (BUN/2.8) + (ethylene glycol/6.2)
      vi. Anion gap:
      $Na - (Cl + HCO_3)$
3. Methanol and ethylene glycol serum concentrations may be monitored to determine severity and to guide therapy in conjunction with an anion gap metabolic acidosis. Often, the ability to obtain these serum concentrations is not readily available and may take several hours to perform; therefore, therapy should not be delayed for lab results.

C. Treatment
1. Treatment is focused on blocking the toxic alcohol metabolism and allowing it to be excreted unchanged in the urine. Treatment is recommended when serum methanol or ethylene glycol concentrations exceed 20 mg/dL, the patient has a documented history of ingestion, there is a high clinical suspicion of ingestion combined with an osmolal gap >10 mOsm/kg, or there is a presence of an anion gap metabolic acidosis of unknown etiology.
2. Gastric decontamination is not recommended.
3. Fomepizole is the preferred antidote because of its predictable response, ease of dosing, and lack of contraindications to use.
   a. Mechanism of action is competitive inhibition of alcohol dehydrogenase.
   b. Dosing is a 15-mg/kg intravenous bolus (as a loading dose); then 10 mg/kg every 12 hours for four doses; then 15 mg/kg every 12 hours until methanol/ethylene glycol concentrations are less than 20 mg/dL.
   c. After 48 hours, fomepizole induces its own metabolism, requiring dosage increases.
   d. Oral administration is effective and may be considered if intravenous access cannot be established (Clin Toxicol 2008;46:181-6).

e. Therapy is discontinued when methanol/ethylene glycol concentrations are less than 20 mg/dL. If the patient is still symptomatic with a normal pH, further workup is warranted, and hemodialysis may be indicated.
f. Hemodialysis increases the clearance of fomepizole; therefore, doses must be administered every 4 hours during hemodialysis. Continuous renal replacement therapy has also been found to increase the clearance of fomepizole, although guidance regarding dosing adjustments is limited and requires further investigation.
g. Adverse effects may include headache, nausea, dizziness, abdominal pain, hypotension, and bradycardia.

4. Ethanol may be administered by diluting 95% alcohol for intravenous, oral, or per-tube administration.
   a. Mechanism of action is competitive inhibition of alcohol dehydrogenase.
   b. Alcohol dehydrogenase has a higher affinity for ethanol.
   c. Intravenous alcohol preparations are no longer commercially available and must be compounded.
   d. Initial dosing is 600–700 mg (7.6–8.9 mL/kg) of a 10% solution, followed by an infusion of 66 mg/kg/hour (0.83 mL/kg/hour). The infusion dose may be initiated at 154 mg/kg/hour (1.96 mL/kg/hour) in chronic drinkers. The goal is to maintain a serum ethanol concentration of 100 mg/dL (0.1%) until symptoms have diminished and methanol or ethylene glycol serum concentrations are undetectable. (Ann Emerg Med 2009;53:439-50).
   e. Disadvantages include frequent monitoring and ICU admission in some institutions.
   f. Adverse effects include CNS depression, nausea, vomiting, abdominal pain, polyuria, and hypoglycemia (especially in children).
5. Hemodialysis should be considered if the clinical condition deteriorates, as evidenced by:
   a. Methanol/ethylene glycol concentration greater than 50 mg/dL
   b. Significant metabolic acidosis
   c. Development of acute renal failure (ethylene glycol) or visual disturbances (methanol)
   d. Development of significant electrolyte abnormalities
6. Additional therapies
   a. Pyridoxine and thiamine
      i. Serve as cofactors in the metabolism of the toxic metabolites of ethylene glycol to nontoxic metabolites
      ii. Pyridoxine promotes the metabolism of glyoxylate to glycine.
         (a) Dosing is generally recommended to be one or more intravenous doses of 50 mg
      iii. Thiamine promotes the metabolism of glycolic acid to a nontoxic metabolite and is also used to prevent or treat Wernicke-Korsakoff syndrome.
         (a) Dosing is generally recommended as 100 mg intravenously, although a concomitant history of chronic alcohol use may warrant use of higher doses (e.g., 500 mg).
   b. Folinic acid
      i. Serves as a cofactor in the metabolism of the toxic metabolites of methanol to nontoxic metabolites. May reduce formate accumulation and reduce the development of metabolic acidosis (Crit Care Clin 2012;28:661-771).
         (a) Dosing is generally recommended to be 1 mg/kg (maximum of 50 mg) intravenously at 4-hour intervals
      ii. Folic acid may be used if folinic acid is unavailable.
   c. Dextrose
      i. Recommended to check a point-of-care level blood glucose concentration before administration (Crit Care Clin 2012;28: 661-771)
      ii. Administer 50 mL of 50% dextrose in water if blood glucose is 70 mg/dL or less or if testing is unavailable.

d. Magnesium – Recommended to administer 1–2 g intravenously for hypomagnesemia (more common in chronic alcohol use)
e. Antiseizure medications
    i. Benzodiazepines are the preferred emergent agents to treat seizures.
    ii. Subsequent urgent control therapy can be implemented with intravenous fosphenytoin, valproate sodium, phenobarbital, or levetiracetam.
        (a) Phenytoin would not be recommended for patients with a history of chronic ethanol ingestion due to enhanced phenytoin metabolism (Gastroenterology 1969;56:412).

D. Monitoring
1. Patient should be closely monitored for resolution of clinical symptoms and return to baseline mental status.
2. Monitor serum electrolytes and blood glucose periodically.
3. Arterial blood gases with a goal of pH greater than 7.2
4. Methanol/ethylene glycol concentrations with a goal of less than 20 mg/dL
5. Monitor the osmolar and anion gaps to ensure the toxic alcohol and metabolites are being cleared appropriately.

Patient Cases

5. A 35-year-old man is admitted to the ED appearing inebriated. He is alert but oriented only to person. His vital signs are BP 122/80 mm Hg, HR 82 beats/minute, and RR 25 breaths/minute. His serum ethanol concentration is 20 mg/dL, and his ethylene glycol concentration is 100 mg/dL. Which is the most appropriate therapy at this time?

A. Fomepizole
B. Ethanol infusion
C. Thiamine
D. Activated charcoal

6. A patient with methanol intoxication is initiated on fomepizole treatment together with hemodialysis. After the 15-mg/kg bolus dose is given, which would be best for adjusting the maintenance fomepizole doses during dialysis?

A. 10 mg/kg every 12 hours
B. 20 mg/kg every 12 hours
C. 10 mg/kg every 4 hours
D. 20 mg/kg every 4 hours

IX. ALCOHOL WITHDRAWAL

A. Background
   1. Alcohol withdrawal is a relatively common consequence of hospital admission.
   2. The strongest risk factor is a history of alcohol withdrawal.
      a. Additional tools have been developed, such as the Prediction of Alcohol Withdrawal Severity Scale, to help identify patients at risk for alcohol withdrawal before development of severe symptoms (Alcohol 2014;48:375-90).

B. Clinical Presentation
   1. Withdrawal symptoms typically occur within 8 hours after blood alcohol concentrations decrease, peak at 72 hours, and are markedly reduced at 5–7 days (N Engl J Med 2014;371:2109-13).
   2. Common symptoms include tremors, diaphoresis, nausea, vomiting, and abnormal vital signs, including hypertension, tachycardia, hyperthermia, and tachypnea (Am J Emerg Med 2013;31:734-42).
   3. Additional symptoms categorized as moderate to severe withdrawal include:
      a. Alcoholic hallucinations – Auditory, visual, or tactile; may last up to 6 days
      b. Alcohol withdrawal seizures (tonic-clonic) – Occur within 72 hours; however, it is highest in the first 24 hours after the last drink.
      c. Delirium tremens – Severe and potentially life-threatening symptom that may develop within 72 hours; presentation includes autonomic hyperactivity, confusion, delirium, psychosis, hallucinations, and seizures.

C. Treatment
   1. The goal of therapy is to keep the patient safe, alleviate and prevent the progression of symptoms, and treat comorbidities (Crit Care Med 2010;38(suppl):S494-S501). Agents for the treatment of alcohol withdrawal are listed in Table 5.
   2. Benzodiazepines are the primary agents used in treatment; they bind to the γ-aminobutyric acid ($GABA_A$) receptor, resulting in hyperpolarization and membrane stabilization.
      a. Lorazepam and diazepam are preferred because of their more predictable effects.
      b. Chlordiazepoxide is not recommended in the acute stabilization setting.
      c. Symptom-triggered therapy is preferred because it reduces benzodiazepine use, duration of mechanical ventilation, and duration of ICU stay.
         i. A front-loading regimen (i.e., a single dose of preventative medication) is recommended for patients at high risk for severe withdrawal (J Addict Med 2020;14:1-72).
      d. Scheduled treatment may be necessary if symptoms are severe or difficult to control.
   3. Use of ethanol to control alcohol withdrawal is controversial and is not routinely recommended.
   4. Phenobarbital
      a. Barbiturate with sedative, hypnotic, and antiseizure activity. Increases the binding of GABA to $GABA_A$ receptor and prolonging the chloride channel opening. It may also concomitantly mitigate neuronal stimulation through NMDA receptor antagonism. Potential advantage over benzodiazepines in alcohol withdrawal because it does not require GABA to be effective (GABA may be depleted), and a longer duration of action.
      b. Historically considered a second-line agent if benzodiazepines fail to adequately control symptoms. Current data suggests that phenobarbital appears to have equivalent outcomes as that of benzodiazepines, and may be more advantageous for certain patients based on patient-specific factors (Pourmand A, AlRemeithi R, Kartiko S, Bronstein D, Tran QK. Evaluation of phenobarbital based approach in treating patient with alcohol withdrawal syndrome: A systematic review and meta-analysis. Am J Emerg Med. 2023 Jul;69:65-75. doi: 10.1016/j.ajem.2023.04.002. Epub 2023 Apr 5. PMID: 37060631.)

   c. May increase the efficacy of benzodiazepines when used in combination by increasing the binding to the $GABA_A$ receptor.
5. Clonidine: α_2-receptor agonist that helps control the catecholamine surge associated with withdrawal (responsible for elevations in BP and HR); however, it will not prevent seizures.
6. Baclofen: Selective $GABA_B$ receptor agonist that reduces the signs and symptoms of alcohol withdrawal; however, it will not prevent seizures.
7. Gabapentin: Increases GABA syntheses and concentrations; may reduce benzodiazepine requirements; however, it may not effectively control seizures when used as monotherapy (Ann Pharmacother 2015;49:897-906).
8. Propofol
   a. General anesthetic acting through $GABA_A$ receptor agonism and N-methyl-D-aspartate (NMDA) receptor antagonism; chronic alcohol use is associated with an up-regulation of NMDA.
   b. Useful for controlling delirium and preventing seizures; patients must be intubated to receive propofol for this indication
9. Dexmedetomidine
   a. α_2-receptor agonist, which may help control BP and HR; however, it will not prevent seizures.
   b. May reduce overall benzodiazepine requirements; however, it lacks the activity necessary to prevent seizures.
   c. Recommended when clonidine cannot be administered and as adjunct therapy
10. Ketamine: NMDA antagonist; may reduce overall benzodiazepine use and can be considered in patients refractory to other therapies (Crit Care Med 2018;46:e768-71).
11. Supportive care: Alcohol-dependent patients are often nutritionally deficient and at risk of Wernicke encephalopathy and hypomagnesemia. High dose intravenous folic acid, thiamine, and magnesium should be administered to nutrient-deficient patients. Dextrose containing fluids may also be considered (Crit Care Med 2016;44:1545-52).

D. Monitoring
1. Clinical Institute Withdrawal Assessment for Alcohol Scale (revised version) (CIWA-Ar) or the Minnesota Detoxification Scale (MINDS) to determine the severity of symptoms and treatment
2. A sedation score, such as the Richmond Agitation-Sedation Scale, should be used in place of the CIWA in intubated patients to guide benzodiazepine dosing.
3. Vital signs every 2–4 hours
4. Electroencephalogram for sustained seizure-related activity

Table 5. Agents for Treatment of Alcohol Withdrawal

Agent	Suggested Starting Dose	Suggested Interval/Infusion Dose Range
Diazepam	5–20 mg PO/IV	Every 6–8 hr
Lorazepam	2–4 mg PO/IV	Every 4–6 hr
Phenobarbital	65– 260 mg or 10 mg/kg IV	Every 15–20 min until symptoms are controlled
Clonidine	0.1–0.3 mg PO	Every 8–12 hr
Baclofen	5–10 mg PO	Every 8–12 hr
Gabapentin	600–800 mg PO	Every 8 hr
Propofol	10–20 mcg/kg/min IV	20–70 mcg/kg/min
Dexmedetomidine	0.2–0.4 mcg/kg/hr IV	0.4–1.5 mcg/kg/hr
Ketamine	0.15–0.2 mg/kg/hr IV	0.2–0.3 mg/kg/hr
Thiamine	100–500 mg IV	Every 8–24 hr for 3–5 days
Folic acid	1–5 mg IV	Once daily for 3–5 days
Magnesium	1–4 g IV	Once daily for 3–5 days

IV = intravenously; PO = orally or per tube.

X. β-BLOCKERS AND CALCIUM CHANNEL BLOCKERS

A. Background
   1. Cardiovascular agents accounted for a little more than 110,000 toxic exposures in 2022 and were a leading cause of death secondary to pharmaceutical exposure.
   2. Two of the most common cardiovascular agents involved in single-agent toxic exposures were β-blockers (28,547 cases and 23 deaths in 2022) and calcium channel blockers (15,718 cases and 30 deaths in 2022).

B. Clinical Presentation
   1. β-Blocker overdoses are characterized by hypotension, bradycardia, and prolonged atrioventricular conduction.
   2. Calcium channel blocker overdoses with the non-dihydropyridine agents are characterized by hypotension, prolonged atrioventricular conduction, bradycardia, lethargy, hyperglycemia, and depressed consciousness. The dihydropyridine agents act peripherally and are primarily associated with vasodilation, hypotension, and reflex tachycardia. However, this receptor selectivity is lost at high doses for both categories (Ann Emerg Med 1993;22:196-200).

C. Treatment
   1. Consider gastric lavage or activated charcoal if patients present within 1–2 hours of overdose. Whole bowel irrigation is recommended for delayed presentation or for sustained- or extended-release formulations.
   2. Maintenance of hemodynamic stability
      a. Goal of therapy is a mean arterial pressure greater than 65 mm Hg or systolic blood pressure (SBP) greater than 90 mm Hg.
      b. Administer isotonic fluids (0.9% sodium chloride or lactated Ringer solution) at 20–30 mL/kg (preferred) or colloidal solutions (e.g., albumin 5% 250 mL).
   3. Administer intravenous calcium chloride or calcium gluconate (first line therapy for calcium channel blocker toxicity).
      a. Calcium chloride 1–2 g (central line is preferred; however, bolus doses may be administered in a peripheral line if needed)
      b. Calcium gluconate 3–6 g. Repeat dose may be given every 15–30 minutes; may also consider continuous infusion at 0.3–0.7 mEq/kg/hour.
   4. Consider sodium bicarbonate for QRS widening, severe acidosis, or dysrhythmia: 1– 2 mEq/kg bolus; may consider a continuous infusion if symptoms persist.
   5. Treat symptomatic bradycardia and hypotension as follows:
      a. Atropine 0.5–1 mg intravenously; if no response, continue to the following options:
      b. Glucagon 5–10 mg (100 mcg/kg) intravenous push for 1 minute (consider for β-blocker toxicity only). If symptom response is achieved from the bolus, may consider a continuous intravenous infusion initiated at the same rate (milligrams per hour) as the bolus dose that achieved response.
         i. Stimulates adenylate cyclase, which increases intracellular cAMP (cyclic adenosine monophosphate), leading to increased inotropy, chronotropy, and cardiac conduction
         ii. Adverse effects include nausea, vomiting, and hyperglycemia. Consider premedicating with an antiemetic intravenously before bolus dosing of glucagon to decrease the risk of nausea and vomiting.
         iii. Use with caution with decreased mental status because of possible aspiration or airway obstruction (also causes lower esophageal sphincter relaxation).
         iv. Not recommended as a preferred treatment option for calcium channel blockers because of limited efficacy and cost

c. Norepinephrine continuous infusion initiated at 2–5 mcg/minute (0.1 mcg/kg/minute) and titrated to target blood pressure in the setting of vasodilatory shock
d. In the presence of cardiogenic shock, epinephrine and dobutamine are the agents of choice. Epinephrine continuous infusion should be initiated at 1 mcg/minute (0.01–0.1 mcg/kg/minute) and titrated to target hemodynamic parameters. Dobutamine continuous infusion should be initiated at 2–20 mcg/kg/min and titrated to target hemodynamic parameters. Dopamine is not recommended in this setting.
e. Transcutaneous or transvenous pacing or intra-aortic balloon pumps may be required in refractory cases

6. Hyperinsulinemic euglycemic therapy (HIET)
   a. Mechanism of action:
      i. Insulin increases the plasma concentrations of ionized calcium, improves the hyperglycemic acidotic state, improves the myocardial use of carbohydrates, and exerts an independent inotropic effect (Am J Crit Care Med 2007;16:498-503).
      ii. Dextrose prevents the development of hypoglycemia after insulin administration.
      iii. Potassium prevents the development of hypokalemia after insulin administration.
      iv. Onset of action is as soon as 5 minutes; however, it may take up to 30 minutes for full effects to be seen.
   b. HIET administration (Clin Toxicol 2011;49:277-83; Am J Health Syst Pharm 2006;63:1828-35; Circulation 2023;148:e149-84):
      i. If baseline glucose is less than 200 mg/dL, administer 50 mL of 50% dextrose in water before insulin administration; may consider an infusion of 10%–20% dextrose to maintain a serum glucose concentration greater than 100 mg/dL
      ii. Correct hypokalemia before initiating insulin therapy. Maintain normokalemia with potassium replacement during insulin infusion and after withdrawal.
      iii. Bolus 1 unit/kg of regular insulin intravenously, followed by a continuous intravenous infusion at 0.5–1 unit/kg/hour; increase rate every 10 minutes to a maximum of 10 units/kg/hour.
         (a) HIET should be continued until existing vasopressor therapy is weaned and hemodynamic stability is obtained. Then, HIET should be tapered over several hours while monitoring for hemodynamic deterioration. Dextrose therapy likely must be continued after discontinuation of HIET to prevent hypoglycemia.
   c. Adverse effects: Hypoglycemia, hypomagnesemia, and hypokalemia
   d. Monitoring:
      i. Vital signs every 15–60 minutes with a goal mean arterial pressure greater than 65 mm Hg and a HR greater than 50 beats/minute
      ii. Serum glucose every 15 minutes; then every 30–60 minutes once stable to target serum concentrations greater than 100 mg/dL
      iii. Serum potassium every hour during the insulin infusion; then every 6 hours to maintain concentrations above 2.8 mEq/L
      iv. Monitor for fluid overload and iatrogenic hyponatremia. Utilization of more concentrated dextrose infusions (e.g., 50% dextrose) and more concentrated insulin infusions (e.g., 16 units/mL) may be necessary.

7. Intravenous lipid emulsion (J Emerg Med 2015;48:387-97; J Med Toxicol 2017;13:124-5)
   a. Mechanism of action is not well known; however, it is thought to be owing to a combination of binding lipid-soluble agents and the provision of free fatty acids that increase cardiac energy and intracellular calcium.
   b. Improves HR and may reduce mortality as an individual treatment or in combination with other therapies.
   c. Evidence is limited to animal models and human case reports, and its role in therapy is controversial.

d. Administration:
   i. Bolus of 1.5 mL/kg of 20% lipid emulsion (such as Intralipid) over 2-3 minutes (typical dose is usually 100 mL)
   ii. A repeat bolus may be considered if no response after initial bolus
   iii. The bolus may be followed immediately by an infusion of 20% lipid emulsion at a rate of 0.25 mL/kg/minute. After 3 minutes at this infusion rate, the infusion rate may be adjusted to 0.025 mL/kg/minute (i.e., 1/10 the initial rate) if there has been a significant response; total duration is not to exceed 6 hours.
   iv. BP, HR, and other available hemodynamic parameters should be recorded at least every 15 minutes during the infusion.
   v. Adverse effects may include pancreatitis, jaundice, coagulopathies, interference with laboratory results, and fat embolism.
   vi. Drug interactions are not well known.

Patient Case

Questions 7 and 8 pertain to the following case.
A 52-year-old man is admitted to the ED with concerns about dizziness and headache. His vital signs are as follows: temperature 98.9°F (37.2°C), BP 87/50 mm Hg, and HR 58 beats/minute. His wife reports that he has a history of hypertension and that he was recently given a diagnosis of being in the early stages of Alzheimer disease. She has brought his medications with her; the 1-month supply was refilled 2 days ago: a bottle of diltiazem CD 120 mg/day (7 tablets remaining) and a bottle of donepezil 5 mg once daily (28 tablets remaining).

7. Which decontamination strategy would provide the most benefit?
   A. Charcoal 25 g every hour until his BP improves
   B. Ipecac 30 mL, followed by 240 mL of water
   C. Polyethylene glycol-electrolyte solution 1500 mL/hour until the rectal effluent is clear
   D. Magnesium citrate 240 mL, followed by 240 mL of water

8. Which antidote would be best to administer first?
   A. Calcium chloride 1 g intravenously over 1 minute
   B. Glucagon 5 mg intravenously over 1 minute
   C. Atropine 2 mg intravenously over 1 minute
   D. Epinephrine 1 mg intravenously over 1 minute

XI. DIGOXIN

A. Background
   1. The cardiac glycosides accounted for 1051 single-agent toxic exposures and 29 deaths in 2022.
   2. Mechanism of action is inhibition of the sodium-potassium adenosine triphosphatase pump and suppression of the atrioventricular node.
   3. Because of its narrow therapeutic index, toxicity has been reported in as many as 35% of patients receiving digoxin (Postgrad Med 1993;69:337-9).

a. The normal therapeutic range is 0.8–2.1 ng/mL.
b. Toxicity may be related to an acute ingestion or may be an issue with chronic use in renal dysfunction.

4. Risk factors for digoxin toxicity include renal failure, advanced age, ischemic heart disease, left ventricular dysfunction, electrolyte imbalances (hypokalemia, hypomagnesemia, hypercalcemia), and hypothyroidism (Postgrad Med 1993;69:337-9).

B. Clinical Presentation
1. Cardiac effects associated with digoxin toxicity include second- and third-degree heart block, tachyarrhythmias, and bradyarrhythmias. More specific examples include fascicular tachycardia, ventricular bigeminy, and ventricular tachycardia (Am J Cardiol 1992;69:108G-119G).
2. Noncardiac effects associated with digoxin toxicity include nausea and vomiting, lethargy, headaches, confusion, and visual disturbances.

C. Treatment
1. Consider decontamination strategies if patients present within 2 hours of overdose.
a. Multidose activated charcoal is beneficial because of the enterohepatic recirculation of digoxin. Load 50–100 g; then 10 g/hour, 10–20 g every 2 hours, or 40 g every 4 hours (Postgrad Med 1993;69:337-9).
b. Colestipol or cholestyramine is an effective drug-binding alternative to charcoal, but it may not be useful in acute toxicity (Am J Cardiol 1992;69:108G-119G).
c. Hemodialysis is not considered effective.
2. Correct serum electrolyte abnormalities.
a. Correct serum potassium concentration to a goal of 3.5–4 mEq/L.
b. Correct serum magnesium concentration to a goal of 1.5–2.2 mg/dL.
c. Correct serum calcium concentration to a goal of 8.5–10.5 mg/dL.
3. Treat symptomatic bradyarrhythmias with atropine 0.5 mg intravenously.
4. Digoxin immune antigen-binding fragments (Fab)
a. Antibodies that bind to digoxin molecules that are then renally excreted
b. Indications for use in acute intoxications (Crit Care Clin 2012;28:527-35; Circulation 2023;148:e149-84):
i. Life-threatening arrhythmias: asystole, ventricular fibrillation or tachycardia, complete heart block, symptomatic bradycardia
ii. Evidence of end-organ damage (e.g., renal failure, altered mental status)
iii. Hyperkalemia (greater than 5–5.5 mEq/L)
c. Products:
i. Digibind (Digoxin Immune Fab): 38 mg per vial
ii. DigiFab (Digoxin Immune Fab): 40 mg per vial
d. Dosing:
i. If amount of digoxin ingested is unknown: 10–20 vials for acute toxicity or 6 vials for chronic toxicity
ii. If the amount of digoxin ingested is known:
Dose (vials) = total body load (0.8 × mg of digoxin ingested)/0.5
iii. If digoxin concentration is known:
Dose (vials) = [serum digoxin concentration (ng/mL) × weight (kg)]/100
iv. May consider lower doses of 1 or 2 vials (40–80 mg) for acute ingestions with repeat doses if necessary
e. Adverse effects include heart failure exacerbation, atrial fibrillation, orthostatic hypotension, hypokalemia, and phlebitis.

D. Monitoring
   1. Monitor vital signs every 30–60 minutes initially. Goal HR of greater than 60 beats/minute and asymptomatic
   2. Monitor serum potassium concentrations hourly for at least the first 6 hours.
   3. Additional serum digoxin concentrations are not recommended after the administration of digoxin immune Fab. Minimal change in serum concentrations will be expected as the digoxin laboratory test will still measure the bound, inactive digoxin molecules. Fab-digoxin complexes may stay in the serum for several days. Repeat serum digoxin concentrations may be checked 24 hours after the initial treatment if Fab is not administered.

XII. ANTIDEPRESSANTS

A. Background
   1. Antidepressants accounted for more than 63,974 single agent toxic exposures and 56 deaths in 2022.
   2. Some of the most common agents involved in toxic exposures were the selective serotonin reuptake inhibitors (SSRIs) and the tricyclic antidepressants (TCAs).
   3. SSRIs block the reuptake of serotonin at the presynaptic neuron.
   4. Patients with SSRI overdoses are often asymptomatic with self-limiting effects (Emerg Med Clin North Am 2007;25:477-97). The most common adverse effects may include drowsiness, tremor, altered mental status, nausea and vomiting, tachycardia, hypotension, seizures, and QRS- or QT-interval prolongation.
   5. TCAs exert many effects, including blocking the reuptake of norepinephrine and serotonin at the presynaptic neuron, blocking muscarinic cholinergic receptors, blocking antihistamine effect, blocking the sodium channel, and, to a lesser degree, blocking α-adrenergic receptors.
   6. Individuals with TCA overdoses may present with the following (Emerg Med Clin North Am 1994;12:533-47):
      a. Cardiovascular: Hypo- or hypertension, tachy- or bradycardia, increased QRS or QT interval, atrioventricular-conduction block, complete heart block
      b. Respiratory: Hypoventilation, crackles, hypoxia
      c. Neurologic: Delirium, lethargy, seizures, coma
      d. Other: Hyperthermia, dry mucous membranes, urinary retention, blurred vision

B. Treatment
   1. There are no specific antidotes for antidepressant overdoses; general supportive care is recommended, with a focus on ABC.
   2. Gastric decontamination is not typically recommended; however, single-dose activated charcoal may be administered within the first hour of exposure (Emerg Med Clin North Am 2000;18:637-54).
   3. Administer crystalloid or colloid fluids to maintain BP and HR, with the goal of a mean arterial pressure >65 mmHg, a SBP greater than 90 mm Hg, and a HR greater than 60 beats/minute.
      a. Norepinephrine or epinephrine may be used if fluid resuscitation alone is unsuccessful.
      b. Dopamine may not be an effective agent because endogenous norepinephrine stores are depleted in an overdose.
   4. Sodium channel blockade
      a. Alkalinization of blood to a pH of 7.45–7.55 is recommended for the TCAs to resolve metabolic acidosis and improve cardiac symptoms. Requires frequent monitoring of arterial pH (varies by effect, but as often as every 15–30 minutes). Sodium load may overcome TCA blockade of sodium channels by increasing the electrochemical gradient.

b. Administer sodium bicarbonate
   i. Recommended bolus dose of 1 mEq/kg (minimum: 50 mEq) intravenously
   ii. May repeat bolus every 15 minutes until ECG stabilized or arterial pH goal achieved.
   iii. May consider a continuous infusion of hypertonic saline in patients with refractory cardiac conduction despite optimal serum alkalinization by increasing sodium load.
c. Proposed indications for sodium bicarbonate include the following (Chest 2008;133:1006-13; Circulation 2023;148:e149-84):
   i. QRS greater than 100–120 milliseconds
   ii. Wide complex tachycardia
   iii. Cardiac arrest
   iv. Right bundle-branch block
   v. Refractory hypotension
d. Replace serum electrolytes, especially magnesium and potassium, if QT prolongation is present. Magnesium replacement is recommended for membrane stabilization in patients with a QTc greater than 500 milliseconds and normal serum magnesium concentrations.
e. Seizures should be managed with benzodiazepines. Phenobarbital and propofol may be considered if the patient is refractory to benzodiazepines and has a stable BP.
   i. Phenytoin and lacosamide are not recommended because of their effects on sodium channels and cardiac adverse effects, respectively.
f. Intravenous fat emulsion
   i. Many case reports for use in amitriptyline and SSRI overdose
   ii. Administration (use lean body mass):
      (a) Bolus of 1.5 mL/kg of 20% lipid emulsion (Intralipid) over 1–5 minutes (typical dose is usually 100 mL for patients weighing 70 kg and more)
      (b) May repeat up to two times for persistent cardiovascular collapse
      (c) Bolus doses should be followed by an intravenous infusion of 0.25–0.5 mL/kg/minute for 60 minutes (typical dose is 18 mL/minute for a 70 kg patient)
      (d) Continue for up to 10 minutes after cardiovascular recovery.

C. Monitoring – Patients should be monitored for clinical improvement for at least 6–8 hours and for a minimum of 24 hours for more severe adverse effects or with citalopram or escitalopram (because of the longer half-lives of these agents).
   a. Monitor for cardiac toxicity with a 12-lead ECG, CK-MB, troponins, BP, and HR.
   b. Monitor for signs and symptoms of respiratory depression with RR and pulse oximetry.

D. Serotonin Syndrome
   1. Excessive serotonin concentrations lead to overstimulation of serotonin-1A and serotonin-2A receptors in the central and peripheral nervous systems (Emerg Med Clin North Am 2007;25:477-97).
   2. Adverse effects include altered mental status, autonomic instability (hyperthermia, tachycardia, hypertension, arrhythmias), and neuromuscular changes (hyperreflexia, increased rigidity).
   3. Diagnosis is made according to clinical findings; many clinicians support the use of the Hunter Serotonin Toxicity Criteria (QJM 2003;96:635-42). By this method, patients are likely to have serotonin toxicity if they have taken a serotonergic agent and one of the following criteria are present:
      a. Spontaneous clonus
      b. Inducible clonus PLUS agitation or diaphoresis
      c. Ocular clonus PLUS agitation or diaphoresis
      d. Tremor PLUS hyperreflexia
      e. Hypertonia PLUS temperature above 100.4°F (38°C) PLUS ocular clonus or inducible clonus

4. Treatment should focus on supportive care with intravenous fluids; symptoms typically resolve within 24–48 hours.
    a. Discontinue the offending agent.
    b. Benzodiazepines should be administered as first line for agitation and muscle rigidity.
    c. Cyproheptadine is a histamine-1 receptor antagonist and nonspecific serotonin receptor antagonist. A single dose of 8–12 mg by mouth should be administered for agitation and muscle rigidity as an adjunct to benzodiazepines. A second dose may be administered in 6–8 hours if symptoms persist.
    d. If condition worsens, may require intubation with continuous infusion benzodiazepines
    e. Although not well studied, case reports have shown the efficacy of dexmedetomidine at doses of 0.05–0.8 mcg/kg/hour (in pediatric patients).

Patient Case

9. A 21-year-old man is admitted to the ED after taking 30 citalopram 20-mg tablets about 2 hours ago. His vital signs are as follows: BP 125/85 mm Hg, HR 77 beats/minute, RR 15 breaths/minute, and temperature 98.7°F (37.1°C). Which is the best intervention for this patient?
    A. Administer lorazepam 2 mg intravenously to prevent seizure activity.
    B. Administer cyproheptadine 8 mg by mouth to prevent muscle rigidity.
    C. Recommend a cooling blanket to prevent serotonin syndrome–related hyperthermia.
    D. Order a 12-lead ECG to monitor for cardiac conduction disturbances.

XIII. ATYPICAL ANTIPSYCHOTICS

A. Background
    1. The atypical antipsychotic agents accounted for about 16,541 toxic exposures and 17 deaths in 2022.
    2. These agents are classified primarily as having D2-dopaminergic receptor and serotonin-2A receptor antagonism. Additional effects include antagonism of the α_1- and histamine-1 receptors.
    3. Adverse effects associated with the atypical antipsychotics are typically self-limiting.
        a. More severe adverse effects may include CNS depression, tachycardia, hypotension, and QT prolongation.
        b. Less severe adverse effects include dizziness, drowsiness, miosis, blurred vision, urinary retention, and CNS excitation.

B. Treatment
    1. There are no specific antidotes for the atypical antipsychotics; general supportive care is recommended, focusing on ABC.
    2. Gastric decontamination is not typically recommended; however, single-dose activated charcoal may be administered within the first hour of exposure if no contraindications exist (J Emerg Med 2012;43:906-13).
    3. Administer crystalloid to maintain BP with a goal mean arterial blood pressure >65 mm Hg, an SBP greater than 90 mm Hg, and an HR greater than 60 beats/minute.
        a. Consider vasopressors if fluid resuscitation is inadequate.
        b. Because of the α-receptor antagonist activity of these agents, norepinephrine or phenylephrine is preferred if vasopressors are needed.
    4. Administer sodium bicarbonate if QRS prolongation is present (quetiapine overdose only)

5. Replace serum electrolytes, especially magnesium and potassium, if QT prolongation. Magnesium replacement is recommended for membrane stabilization in patients with a QTc greater than 500 milliseconds and normal serum magnesium concentrations.
6. Seizure activity should be managed with benzodiazepines, barbiturates, or propofol.
7. Anticholinergic symptoms can potentially be treated with IV physostigmine in select patients (e.g., normal ECG); however, substantial controversy exists with this approach, particularly in the setting of a concomitant TCA overdose and its overall functional use considering its short half-life (Ann Emerg Med 2000;35:374-81). Currently, it is not frequently used and further investigations into its safety and efficacy are necessary (J Med Toxicol 2015;11:179-84).
8. Lipid emulsion therapy may be effective because of the high lipophilicity of these agents and may be considered if more traditional treatment means do not improve the cardiovascular complications of decreased HR and/or BP (J Emerg Med 2012;43:906-13). Administer an intravenous bolus dose of 1.5 mL/kg of 20% lipid emulsion over 2–3 minutes, followed by a continuous intravenous infusion of 0.25–0.5 mL/kg/min for 60 minutes, if necessary.

C. Monitoring: Patients should be monitored for clinical improvement for at least 8–12 hours.
   1. Monitor for cardiac toxicity with a 12-lead ECG, CK-MB, and troponins.
   2. Monitor for respiratory depression with RR and pulse oximetry.

XIV. LITHIUM

A. Background
   1. Lithium was associated with almost 3341 toxic exposures and seven deaths in 2022.
   2. Mechanism of action is through an influence on serotonin and norepinephrine reuptake, inhibition of the phosphatidylinositol cycle, and inhibition of the post-synaptic D2 receptor.
   3. Adverse effects associated with lithium include:
      a. Acute overdose:
         i. GI: Nausea, vomiting, diarrhea
         ii. CNS: Confusion, tremor, myoclonus, seizures, coma
         iii. Cardiovascular: T-wave inversion, ventricular arrhythmias
      b. Chronic adverse effects:
         i. Endocrine: Hypothyroidism, myxedema coma
         ii. Nephrogenic diabetes insipidus

B. Treatment
   1. There are no specific antidotes for lithium; general supportive care is recommended, focusing on ABC.
   2. Gastric decontamination is not typically recommended in toxic acute ingestions. Single-dose activated charcoal is not effective for lithium overdoses; whole bowel irrigation may be beneficial.
   3. Administer crystalloid to maintain BP, with a goal SBP greater than 90 mm Hg. Consider vasopressors if fluid resuscitation is not adequate.
   4. Replace serum electrolytes, especially magnesium and potassium, if QT prolongation is present.
   5. Seizure activity should be managed with benzodiazepines, barbiturates, or propofol.
   6. Lithium overdoses are primarily managed with fluid resuscitation and renal replacement therapy.
      a. Saline infusions may be administered if there are no contraindications to fluid therapy (goal is a serum sodium concentration of 140–145 mEq/L). Lithium clearance is reduced in hyponatremia.
      b. Intermittent hemodialysis may require several sessions to fully remove lithium concentrations because of the rebound of lithium concentrations that occurs after dialysis sessions. Consider continuous renal replacement therapy (CRRT) for hemodynamically unstable patients.

c. Suggested indications for the use of hemodialysis include (Chest 2008;133:1006-13):
   i. Severe toxicity (severe altered mental status or seizures)
   ii. Renal failure (cannot eliminate lithium)
   iii. Lithium concentrations greater than 2.5 mmol/L in chronic exposures
   iv. Lithium concentrations greater than 4 mmol/L in acute exposures

C. Monitoring – Patients should be monitored for clinical improvement for at least 8–12 hours.
   1. Monitor for cardiac toxicity with a 12-lead ECG, CK-MB, and troponins.
   2. Monitor for respiratory depression with RR and pulse oximetry.
   3. Monitor renal function with urine output, BUN, and SCr.
   4. Monitor baseline lithium concentrations and then every 6 hours after until concentrations have decreased to less than 1.5 mmol/L (normal 0.6–1.2 mmol/L).

Patient Case

10. A 24-year-old woman is brought to the ED by her roommate. She has been in a normal state of health, but the roommate is concerned because she "seems really out of it." According to the roommate, the patient had an appointment with the physician today, and she had been given a prescription to refill olanzapine 5 mg by mouth daily, but the bottle is empty. On physical examination, she is alert and oriented person, place and time but she dozes off several times. Her vital signs are stable, and a 12-lead ECG shows sinus tachycardia. Which intervention is most appropriate for this patient?
    A. Lactated Ringer solution 500 mL intravenously
    B. 8.4% sodium bicarbonate 50 mL intravenously
    C. Lorazepam 2 mg intravenously
    D. Clinical monitoring for 6 hours

XV. ORAL HYPOGLYCEMICS

A. Background
   1. Oral hypoglycemics accounted for 5944 single-agent exposures and 29 deaths in 2022.
   2. The most common oral hypoglycemic involved in toxic exposures was metformin, followed by the sulfonylureas and the glucagon-like peptide-1 (GLP-1) receptor agonists.
   3. The most serious adverse effects were reported with the sulfonylureas; however, most fatalities were associated with metformin.

B. Clinical Presentation
   1. Clinical signs and symptoms include hypoglycemia (not with metformin), nausea, vomiting, dizziness, tachycardia, and diaphoresis.
   2. More severe adverse effects include seizures, palpitations, tachyarrhythmias, electrolyte abnormalities, and metabolic (lactic) acidosis.

C. Treatment
   1. Stabilization of the ABC
   2. Identifying the causative agent is important because specific treatment will vary by the agent involved.
   3. Consider gastric decontamination with single-dose activated charcoal if patients present within 1 hour of overdose.
   4. Observe clinically asymptomatic patients for a minimum of 8 hours (Am J Health Syst Pharm 2006;63:929-38).

5. For symptomatic patients or blood glucose less than 70 mg/dL, treat with glucose:
   a. Conscious patients: Administer 8 oz of an oral carbohydrate (such as juice, non-diet sodas, or milk) or oral glucose tablets or gels.
   b. Unconscious patients: Administer intravenous dextrose, 0.5–1 g/kg
   c. Repeat doses may be required; consider a continuous infusion of dextrose if needed. Glucose concentrations should be monitored often (every 15–60 minutes) until stable.
   d. Use caution to avoid overcorrection of serum glucose.
6. Octreotide
   a. Mechanism of action is a somatostatin analog that inhibits the secretion of insulin.
   b. Primarily studied in sulfonylurea overdose, but considered a treatment option for all oral hypoglycemic toxic exposures
   c. Administer 50 mcg subcutaneous or intravenous, followed by 50-mcg doses every 6 hours. Intravenous dextrose infusion should be slowly tapered off.
   d. Adverse effects include headache, dizziness, nausea, abdominal pain, and sinus bradycardia.
7. Sodium bicarbonate
   a. Indicated for severe metformin-associated lactic acidosis
   b. 1–2 mEq/kg or 50–200 mEq of 8.4% sodium bicarbonate intravenously
8. Glucagon
   a. Mechanism of action is stimulation of gluconeogenesis.
   b. May trigger additional insulin secretion, leading to a secondary hypoglycemia
   c. May provide benefit in prehospital settings when oral or intravenous options are not available, but is not routinely recommended
   d. Not recommended in pediatric patients, in malnourished patients, or for sulfonylurea toxic exposures
9. Hemodialysis or continuous renal replacement therapy may be necessary to enhance metformin clearance in severe cases.

D. Monitoring
   1. Regular assessment of vital signs and mental status (Emerg Med J 2006;23:565-7)
   2. Measure capillary blood glucose at a minimum of every hour for 24 hours with a goal of greater than 70 mg/dL.
   3. Measure BP hourly, especially after octreotide administration.

Patient Case

11. As the pharmacist in the ICU satellite, you receive a call from a distressed nurse about a patient in the cardiac step-down unit. The patient was found unconscious, and, on investigation, it was discovered that he had received a glyburide 20-mg tablet 1 hour earlier that was meant for another patient. The patient has stable vital signs, but his point-of-care blood glucose concentration is 37 mg/dL. Which intervention is most appropriate at this time?
    A. 8 oz of milk by mouth
    B. 50 mL of 50% dextrose in water intravenously
    C. Octreotide 100 mcg subcutaneously
    D. Glucagon 1 mg intramuscularly

XVI. DRUGS OF ABUSE

A. Background
   1. Miscellaneous stimulants and street drugs accounted for 46,753 toxic exposures and 186 deaths in 2022.
   2. The most common drugs of abuse involved in toxic exposures were amphetamines, marijuana (and derivatives), cocaine, methamphetamines, and heroin.
   3. Total numbers are difficult to determine because few of these agents can be detected with current techniques.

B. Amphetamines/Methamphetamines/MDMA (ecstasy)
   1. Mechanism of action/toxicity: stimulation of the CNS, peripheral release of catecholamines, inhibition of catecholamine reuptake, and inhibition of monoamine oxidase (Intensive Care Med 2004;30:1526-36)
   2. Clinical presentation (Intensive Care Med 2004;30:1526-36)
      a. Common: Confusion, tremor, anxiety, agitation, irritability, mydriasis, tachyarrhythmias
      b. Severe: Hepatocellular necrosis, acute hepatitis, myocardial ischemia, hypertension, cerebral hemorrhage, seizures, hyponatremia
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration and airway maintenance
      b. Gastric lavage or activated charcoal if within 1 hour of ingestion
      c. BP control with benzodiazepines, α-blocker, or vasodilator (e.g., nitroglycerin, nitroprusside, nicardipine). Avoid the use of β-receptor blocking agents because of risk of unopposed α-receptor activity leading to an increase in BP.
      d. Benzodiazepines for agitation, anxiety, psychosis, and/or uncontrolled hypertension titrated to effect. Haloperidol or phenothiazines may be considered for use (with caution) in patients with primary psychiatric disorders or dopamine-mediated movement disorders.
      e. Cooling therapy if hyperthermia present
      f. Monitor for serotonin syndrome.

C. Synthetic Cannabinoids: K2/Spice
   1. Mechanism of action/toxicity: stimulation of the cannabinoid-1 (CB1) and cannabinoid-2 (CB2) receptors (Pharmacotherapy 2015;35:189-97)
      a. CB1 receptors modulate glutamate and GABA and are found both peripherally and centrally.
      b. CB2 receptors are located in immune tissue and the CNS and modulate pain and emesis.
   2. Clinical presentation (J Pharm Pract 2015;28:50-65)
      a. Predominantly euphoria or excited delirium
      b. Common adverse effects include:
         i. Psychiatric: Agitation, anxiety, hallucinations, paranoia, catatonia
         ii. Neurologic: Cognitive impairment, ataxia, dizziness, headache, seizures
         iii. Cardiovascular: Tachycardia, palpitations, hypertension
         iv. GI: Nausea, vomiting
         v. Renal: Acute kidney injury, rhabdomyolysis
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration
      b. Benzodiazepines for agitation or seizure activity. Antipsychotics for agitation may be considered for use (with caution) in patients with primary psychiatric disorders or dopamine-mediated movement disorders.

D. Cocaine
   1. Mechanism of action/toxicity (Intensive Care Med 2004;30:1526-36):
      a. Potent sympathetic nervous system stimulant
      b. Inhibits the presynaptic reuptake of epinephrine and norepinephrine; also stimulates norepinephrine release
      c. Acts as a reuptake inhibitor of dopamine, norepinephrine, and serotonin
   2. Clinical presentation (Intensive Care Med 2004;30:1526-36)
      a. Predominantly euphoria or excited delirium
      b. Common/severe adverse effects include:
         i. Psychiatric: Agitation, anxiety, psychosis, delirium
         ii. Neurologic: Stroke, subarachnoid or intracranial hemorrhage, seizures
         iii. Cardiovascular: Tachycardia, hypertension, palpitations, arrhythmias, heart failure, aortic dissection
         iv. GI: Gastric ulcers, gastric perforation, bowel ischemia
         v. Respiratory: Status asthmaticus, pulmonary hypertension, pulmonary edema, alveolar hemorrhage
         vi. Renal: Acute kidney injury, rhabdomyolysis
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration
      b. Gastric decontamination is typically not recommended. Activated charcoal may provide benefit if given within 1 hour of oral ingestion.
      c. Benzodiazepines are administered for agitation or seizure activity and titrated to relaxation. Because of their ability to block the CNS stimulant effect of cocaine, benzodiazepines are also an effective treatment for hypertension and tachycardia.
      d. BP control with benzodiazepines, α-blocker, or vasodilator (e.g., nitroglycerin, nitroprusside, nicardipine). Avoid the use of β-receptor blocking agents because of risk of unopposed α-receptor activity leading to an increase in BP.

E. Synthetic Cathinones: Bath Salts
   1. Mechanism of action/toxicity: increases presynaptic concentrations of serotonin, dopamine, and norepinephrine by stimulating release and antagonizing monoamine reuptake (Pharmacotherapy 2014;34:745-57)
   2. Clinical presentation (J Pharm Pract 2015;28:50-65):
      a. Predominantly euphoria, increased energy, and alertness
      b. Common adverse effects include:
         i. Psychiatric: Agitation, aggression, anxiety, hallucinations, paranoia
         ii. Neurologic: Amnesia, confusion, insomnia, seizures, dizziness, headache
         iii. Cardiovascular: Angina, hypertension, tachycardia, palpitations
         iv. GI: Abdominal pain, nausea, vomiting, anorexia, hepatic failure
         v. Renal: Acute kidney injury, increased SCr, and rhabdomyolysis
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration
      b. Benzodiazepines or antipsychotics for agitation or anxiety
         i. Haloperidol is an alternative option, but it should be used with caution and close monitoring because of the potential to worsen hyperthermia.
         ii. Consider propofol or dexmedetomidine for severe symptoms.
      c. Antiemetics for nausea and vomiting
      d. Monitor for serotonin syndrome.

F. Piperazines
   1. Mechanism of action/toxicity: enhances neurotransmitter release and reuptake inhibition of dopamine, serotonin, and norepinephrine release (Emerg Med Clin North Am 2014;32:1-28)
   2. Clinical presentation (J Pharm Pract 2015;28:50-65):
      a. Predominantly euphoria and increased energy
      b. Common adverse effects include:
         i. Psychiatric: Agitation, anxiety, hallucinations, psychosis, depressed mood or mood swings, paranoia
         ii. Neurologic: Confusion, insomnia, tremor, seizures, dizziness, headache
         iii. Cardiovascular: Angina, hypertension, tachycardia, palpitations, QT prolongation
         iv. GI: Abdominal pain, nausea, vomiting
         v. Renal: Urinary retention
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration
      b. Benzodiazepines for agitation or seizures
      c. Avoid antipsychotics because of the risk of worsening hyperthermia, extrapyramidal effects, and hypotension or arrhythmias.
      d. Treat hypertension with parenteral antihypertensives or clonidine.
      e. Hyperthermia treatment if above 104°F (40°C)
      f. Monitor for serotonin syndrome.

G. Ketamine
   1. Mechanism of action/toxicity (Lancet 2005;365:2137-45):
      a. Noncompetitive NMDA receptor antagonist (blocks glutamate and aspartate)
      b. Mild to moderate blockade of catecholamine reuptake
   2. Clinical presentation:
      a. Predominantly hallucinations and vivid dreams
      b. Common adverse effects include:
         i. Psychiatric: Impaired memory, cognitive dysfunction, severe agitation
         ii. Cardiovascular: Hypertension, tachycardia, cardiac arrhythmias
         iii. Respiratory: Laryngospasm, apnea, respiratory depression
         iv. GI: Anorexia, nausea, vomiting
         v. Genitourinary: Cystitis, irritable bladder, urethritis
   3. Treatment
      a. Mostly supportive care with intravenous fluid administration
         i. Monitor for rhabdomyolysis
         ii. Aspiration precautions are recommended in comatose patients
         iii. Urinalysis and serum chemistries if symptomatic for cystitis
      b. Activated charcoal may provide benefit if given within 1 hour of oral ingestion. Additional doses every 4 hours may be considered.
      c. Benzodiazepines are recommended for agitation or seizures.
      d. Haloperidol may be used if benzodiazepines are not effective. Monitor closely because of the potential of lowering the seizure threshold and worsening of dystonia, hypotension, neuroleptic malignant syndrome, and/or myoglobinuria.

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ANSWERS AND EXPLANATION TO PATIENT CASES

1. **Answer: B**

The most important first step in all drug overdose cases is to try to stabilize the patient's ABC (Answer B is correct). This may involve the use of supplemental oxygen or advanced airway management, establishment of intravenous access, and administration of intravenous fluids. Once the patient is stable, the process of identifying the suspected toxin can begin (Answer D is incorrect). This may include thoroughly examining the patient, speaking with family or first responders, and communicating with the patient's physicians and pharmacies. Blood and urine samples may be sent for quantitative or qualitative toxicologic assays (Answer A is incorrect). A coma cocktail may provide some benefit, but a clear cause should be established before considering its use (Answer C is incorrect).

2. **Answer: B**

The patient has symptoms of an acute acetaminophen overdose, and stabilizing the patient, together with providing good supportive care, is indicated until a determination for additional therapy can be made (Answer B is correct). Typical decontamination strategies may provide benefit, but they do not definitively improve patient outcome. Magnesium citrate (and cathartics as a whole) is not considered an effective decontamination strategy (Answer A is incorrect). Gastric lavage is of most benefit within the first 60 minutes of exposure, and the potential adverse effects outweigh any potential benefit (Answer C is incorrect). Similarly, single-dose charcoal requires more rapid administration (Answer D is incorrect).

3. **Answer: C**

The patient is considered at high risk of developing hepatic damage from acetaminophen and requires therapy with intravenous acetylcysteine (Answer D is incorrect). The dose of intravenous acetylcysteine is as follows (doses are calculated using actual body weight): loading dose: 150 mg/kg in 200 mL of 5% dextrose in water for 60 minutes; maintenance dose: 50 mg/kg in 500 mL of 5% dextrose in water for 4 hours, followed by 100 mg/kg in 1000 mL of 5% dextrose in water for 16 hours (Answer C is correct and Answer B is incorrect). Oral dosing of acetylcysteine is not a viable option because of the patient's severe nausea and vomiting (Answer A is incorrect).

4. **Answer: C**

This patient has an acute salicylate overdose. Although his serum salicylate concentrations are in the therapeutic range, he has symptoms consistent with salicylate toxicity, as evidenced by his nausea, tachycardia, and respiratory alkalosis. He is currently stable, but his serum salicylate concentrations may continue to rise; therefore, enhanced elimination with serum bicarbonate is the best option (Answer C is correct). His vital signs, which are stable, should be monitored for changes; however, although he is not experiencing signs of significant dehydration, he would benefit from the fluid administration of sodium bicarbonate. Sodium chloride would be more beneficial if his vital signs were more unstable (Answer A is incorrect). His RR, which is elevated, should be monitored; however, he is alert and able to communicate and therefore does not need intubation at this time (Answer B is incorrect). He is also not indicated for hemodialysis because of his moderate symptoms, but this could be considered if his condition deteriorates (Answer D is incorrect).

5. **Answer: A**

The most appropriate therapy for an ethylene glycol intoxication is fomepizole (Answer A is correct). An ethanol infusion is a possible treatment option, but it is not preferred because of the difficulties in dosing and adverse effects (Answer B is incorrect). Thiamine is a cofactor in the metabolism of ethylene glycol, but it would not be preferred to administer thiamine before fomepizole (Answer C is incorrect). Activated charcoal is not an option for gastric decontamination because it is not effective for alcohols (Answer D is incorrect).

6. **Answer: C**

After the initial bolus of fomepizole, 10 mg/kg should be administered every 12 hours. Because of the increased clearance of fomepizole during hemodialysis, the frequency is changed to every 4 hours during dialysis (Answer C is correct and Answer A is incorrect). When dialysis is completed, the dose returns to 10 mg/kg administered every 12 hours; and once the patient has been on 48 hours of therapy, dosing increases to 15 mg/kg because of self-induction. There is no indication for a dose increase (Answer B and D are incorrect).

7. **Answer: C**
The best treatment option for this patient is whole bowel irrigation because of the extended-release formulation of diltiazem (Answer C is correct). Activated charcoal may provide some benefit, but, similar to ipecac, the time interval is not known, and diltiazem does not undergo enterohepatic recirculation (Answer A is incorrect). Ipecac is not recommended because it may impede treatment with more effective treatment options and because it is no longer manufactured in the United States (Answer B is incorrect). A cathartic would not be useful in this situation; guidelines recommend its use only in combination with other decontamination strategies, not as a single agent (Answer D is incorrect).

8. **Answer: A**
There are several potential antidotes for a calcium channel blocker overdose. Calcium is the most effective, and it should be given by bolus, followed by continuous infusion if needed (Answer A is correct). Glucagon is not an effective antidote and is therefore not an option for this patient (Answer B is incorrect). Atropine is effective for symptomatic bradycardia caused by the calcium channel blocker, but the dose should be 0.5–1 mg (Answer C is incorrect). Epinephrine is an alternative to glucagon, but this dose is excessive for a patient not experiencing cardiac arrest (Answer D is incorrect).

9. **Answer: D**
Most of the SSRIs are relatively safe, and many patients will present as asymptomatic after an overdose. However, there is a potential for a patient to develop serious adverse effects, such as serotonin syndrome, seizures, and cardiac toxicity. Although this patient is stable and has no specific concerns, it is recommended to check a 12-lead ECG to measure for QT-interval prolongation and treat with sodium bicarbonate, if necessary (Answer D is correct). A benzodiazepine should be administered if muscle rigidity develops, but it should not be used as a prophylactic measure (Answer A is incorrect). It is recommended that the patient be observed for at least 6–8 hours. Cyproheptadine is only indicated for symptomatic patients (Answer B is incorrect). Measures should be performed to reduce hyperthermia if a serotonergic syndrome develops, but this should be treated with measures to reduce muscle activity (i.e., sedation or chemical paralysis), not by applying measures to enhance surface cooling (Answer C is incorrect).

10. **Answer: D**
Although the patient appears to have taken an overdose of olanzapine, she is experiencing only mild symptoms. The best intervention would be to monitor her for 6 hours for the progression of her symptoms or development of additional complications (Answer D is correct). Intravenous fluids would be appropriate if the patient has dehydration or hypotension (Answer A is incorrect). Sodium bicarbonate is indicated for QRS prolongation and is not warranted at this time (Answer B is incorrect). Olanzapine does not cause seizures; therefore, lorazepam would not be indicated (Answer C is incorrect).

11. **Answer: B**
The most appropriate intervention at this time is to give the patient intravenous dextrose (Answer B is correct). Oral glucose is a viable option, but it cannot be administered to an unconscious patient without oral access (Answer A is incorrect). Octreotide should be reserved for use if the administration of a glucose solution fails to raise the blood glucose above 70 mg/dL for two consecutive readings (Answer C is incorrect). Glucagon is a potential option for treatment, but because the patient has intravenous access, the intramuscular route would not be preferred (Answer D is incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: C**

The best option for this patient right now is to administer octreotide 50–100 mcg subcutaneously (Answer C is correct). The patient has not responded to two doses of intravenous dextrose, as evidenced by point-of-care glucose concentrations less than 70 mg/dL; therefore, additional doses of dextrose are not indicated (Answer A is incorrect). Although glucagon is also a potential option, it is not recommended for sulfonylurea exposures (Answer B is incorrect). Sodium bicarbonate intravenously may be indicated in this scenario if the patient had metformin-associated lactic acidosis; however, the patient is not on metformin and has no signs of lactic acidosis (Answer D is incorrect).

2. **Answer: A**

Any of the options listed in this question are possible treatments for a patient with a β-blocker overdose who is not responding to the administration of intravenous fluids and calcium. The optimal choice ultimately involves efficacy and appropriate dosing. Glucagon is an option, and it should be dosed at 5–10 mg intravenously initially (Answer A is correct). Atropine is an option for the patient's bradycardia, but the initial recommended dose is 0.5–1 mg intravenously (Answer B is incorrect). Hyperinsulinemic euglycemic therapy may be preferred in this setting; however, the correct bolus dose is 1 unit/kg intravenously (Answer C is incorrect). Dopamine is an option for the treatment of hypotension and bradycardia, but the correct dose would be the initiation of an infusion at 5–10 mcg/kg/minute titrated to effect (Answer D is incorrect).

3. **Answer: B**

Given the patient's presentation and the common toxidromes, the most likely scenario is a cholinergic agent (Answer B is correct). The patient is experiencing bradycardia with a normal BP and RR, has a decrease in mental status, and is experiencing nausea. Although not an absolute, anticholinergics and sympathomimetics are more commonly associated with tachycardia (Answers A and D are incorrect). Similarly, opioids are typically associated with a decrease in respirations (Answer C is incorrect).

4. **Answer: A**

The patient is experiencing QT prolongation after an atypical antipsychotic overdose. It is important to stabilize the patient by administering intravenous sodium bicarbonate and electrolyte replacement (Answer A is correct). Her potassium concentration is low, requiring replacement. Because the time interval of the overdose is not known, there is limited benefit for activated charcoal (Answer B is incorrect). Although her magnesium concentration is normal, it should be monitored; however, her magnesium concentration does not require replacement at this time because her QTc is less than 500 milliseconds (Answer C is incorrect). Lorazepam is not indicated for prophylaxis of seizure activity (Answer D is incorrect).

5. **Answer: A**

This patient has the clinical signs and symptoms of alcohol withdrawal. Management should focus on the patient's safety and controlling his symptoms, and treatment should be administered using a symptom-triggered therapy strategy. The primary agents used to control symptoms are the benzodiazepines, and lorazepam is a good option (Answer A is correct). Barbiturates such as phenobarbital are typically reserved for patients who do not respond to benzodiazepine therapy because of benzodiazepine's long elimination half-life and stronger sedative effects and oral dosing may be difficult with his level of confusion (Answer B is incorrect). Propofol should be avoided in non-intubated patients (Answer C is incorrect). Clonidine is a potential option, especially because this patient has borderline hypertension, but oral dosing may be difficult with his level of confusion (Answer D is incorrect).

6. **Answer: B**

The patient is experiencing an unintended opioid overdose, as evidenced by the decreased RR and decreased consciousness. Administration of the antidote, naloxone, is the best option (Answer B is correct). Because 2 hours have passed since the methadone dose was given, there is limited usefulness for activated charcoal at this time, and it would not be advisable to administer it to an unconscious patient without an established airway (Answer A is incorrect). Whole bowel irrigation is also not useful in this situation because it is too late to prevent

drug absorption together with the airway safety concern (Answer C is incorrect). Administration of intravenous fluids would be beneficial to improve BP but should not be administered in this case before naloxone (Answer D is incorrect).

7. **Answer: D**

The patient is not responding to the initiation of intravenous fluids and calcium gluconate, so HIET is warranted. Because of the patient's low serum potassium concentrations, it is critical to replace this before administering insulin (Answer A is incorrect and Answer D is correct). The patient's glucose concentration is greater than 200 mg/dL, so additional glucose need not be given at this time (Answer B is incorrect). Full effects may take up to 30 minutes to be seen, but this should not prevent the initiation of HIET (Answer C is incorrect).

8. **Answer: B**

The patient is not responding to the initiation of intravenous fluids, calcium, and HIET. The most appropriate option at this time would be to increase the rate of the insulin infusion and initiate vasopressor therapy to improve hemodynamic stability in the interim (Answer B is correct). The initial infusion rate is 0.5–1 unit/kg/hour and is titrated every 15–20 minutes until hemodynamically stable. The next option would be to initiate a vasopressor agent (Answers A and C are incorrect). From the choices listed, the best first option is norepinephrine initiated at 4 mcg/minute and titrated to the desired effect. Epinephrine is also a possible option, but it would be recommended if the patient were not responding to increasing doses of norepinephrine. Intravenous lipid emulsion is a potential therapy, but it is typically administered in a patient with severe decompensation caused by a lipophilic medication who is not responding to fluids or vasopressors (Answer D is incorrect).